Journal of Affective Disorders Reports

Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urban Differences

--Manuscript Draft--

Manuscript Number:	JADR-D-26-00039R1
Full Title:	Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urban Differences
Article Type:	Research Paper
Keywords:	Depression; Decision-making autonomy; Pregnancy loss; Rural-urban disparities; Lesotho; Spatial analysis
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	Maria Bintey Kabir, MBBS
Order of Authors Secondary Information:	
Abstract:	Objectives To assess the association between depressive symptoms and decision-making autonomy with rural urban inequalities, and mapped province-level spatial disparities in depression prevalence in ever-married women in Lesotho. Methods This study used cross-section data of 3,297 women aged 15–49 years from the 2023–24 Lesotho Demographic and Health Survey (LDHS). Depression was measured using the Patient Health Questionnaire-9 (PHQ-9 ≥10). Women's decision-making autonomy was measured using a composite empowerment index based on participation in five household decision-making domains. Survey-weighted multivariable logistic regression models were fitted at the national level and stratified by place of residence. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC). Spatial analyses were conducted to examine district-level variation in depression prevalence and autonomy. Results: Overall, 7.4% (95% CI: 6.3–8.7) of women reported moderate-to-severe depressive symptoms. Women with decision-making autonomy showed lower odds of depression nationally (AOR = 0.63; 95% CI: 0.39–0.98), with a stronger effect in rural areas (AOR = 0.44; 95% CI: 0.27–0.74). Depression prevalence was highest in central–northern districts (Maseru, Berea, Leribe, Butha-Buthe) and lower levels of autonomy highlighted in southern and south-western provinces. Conclusions Women's decision-making autonomy is significantly associated with lower odds of depressive symptoms in Lesotho, particularly among rural women. The detected rural–urban and spatial inequalities highlight the need for context-specific mental health and reproductive health interventions that incorporate women's empowerment as a core component.
Opposed Reviewers:	
Additional Information:	
Question	**Response**

To:
Editor-in-Chief
Journal of Affective Disorders

Subject: Submission of Manuscript – *Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urban Differences*

Dear Editor,

I am pleased to submit our manuscript entitled *"Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urban Differences"* for consideration in *Journal of Affective Disorders*.

This study investigates the association between women's decision-making autonomy and depressive symptoms, focusing on rural–urban disparities, using nationally representative data from the 2023–24 Lesotho Demographic and Health Survey (LDHS). We applied survey-weighted multivariable logistic regression and spatial mapping to identify province-level variations in depression prevalence and autonomy. Our findings show that higher decision-making autonomy is significantly associated with lower odds of depressive symptoms, particularly among rural women, and highlight geographic clusters with the highest mental health burden. These results emphasize the importance of incorporating women's empowerment into context-specific mental health interventions.

We believe this manuscript will be of interest to your readership as it provides novel insights into mental health disparities among women in sub-Saharan Africa, highlights spatial inequalities in depression, and underscores the protective role of decision-making autonomy.

The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. We have no conflicts of interest to declare.

Thank you for considering our submission. We look forward to your review and are happy to provide any additional information if required.

Sincerely,

Md Salek Miah
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ELSEVIER

Authorship change request

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(*continued from the previous page*)

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Part 1. General submission information

To be completed by **the Corresponding Author**.

Manuscript details

Journal title	Journal of Affective Disorders Reports
Manuscript and/ or article number	JADR-D-26-00039
Manuscript title	Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urba

Change(s) requested (check all that apply)

☑ Add new author(s)	☐ Remove author(s)
☐ Change the Corresponding Author	☐ Change the order of authors*

* If ONLY changing order of existing authors, go directly to Part 3. Do not complete Part 2.

Part 2. Indicate author(s) to be added or removed

For each author to be added or removed complete one template below.**

In the "**Reason for this change**" section, please include as much detail as possible so we can evaluate if the change is approved. At a minimum, this should include explanations for both:

- why the change is being requested, and
- why the author was/was not included in the original author list.

Important Note: If this part is either not provided, incomplete, or the reasons provide insufficient detail or do not address the points above, your request will be denied, and your submission may be rejected.

2.1 Author change information

Given/first name(s)	Maria Bintey
Family/last name	Kabir
Email address	mbkabir67@gmail.com
Institution	Shaheed Suhrawardy Medical College, Dhaka-1207, Bangladesh; Study Physician, Projahnmo Research Foundation, Dhaka-1213, Bangladesh

Change(s) requested for this author ☐ Remove author ☑ Add new author ☐ Make Corresponding Author

Author's Individual contributions (required for author additions only) see CRediT Contributor Roles Taxonomy

☐ Conceptualization	☐ Data curation	☐ Formal analysis
☐ Funding acquisition	☐ Investigation	☐ Methodology
☐ Project administration	☐ Resources	☐ Software
☐ Supervision	☑ Validation	☐ Visualization
☐ Writing – original draft	☑ Writing – review & editing	

Reason for this change

The addition of Dr. Maria Bintey Kabir, who has completed MBBS and holds a specialization in Public Health, is requested to strengthen the scientific quality of the revised manuscript in response to the reviewers' comments. Following the major revision decision, Dr. Kabir provided substantial intellectual contributions to the study, including supervising the revision process, refining the theoretical framework, improving the methodological approach, and critically revising the interpretation of results and discussion sections. Dr. Kabir was not included in the original author list as her involvement began after the initial submission. Her contributions during the revision process meet the authorship criteria, and her inclusion is essential to ensure the rigor, clarity, and overall quality of the revised manuscript. All authors, including Dr. Kabir, have reviewed and approved this authorship change and agree with the updated author list.

2.2 Author change information

Given/first name(s)	
Family/last name	
Email address	
Institution	
Change(s) requested for this author	☐ Remove author ☑ Add new author ☐ Make Corresponding Author

Author's Individual contributions (required for author additions only)
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☐ Conceptualization	☐ Data curation	☐ Formal analysis
☐ Funding acquisition	☐ Investigation	☐ Methodology
☐ Project administration	☐ Resources	☐ Software
☐ Supervision	☐ Validation	☐ Visualization
☐ Writing – original draft	☐ Writing – review & editing	

Reason for this change

2.3 Author change information

Given/first name(s)	
Family/last name	
Email address	
Institution	
Change(s) requested for this author	☐ Remove author ☑ Add new author ☐ Make Corresponding Author

Author's Individual contributions (required for author additions only) see [CRediT Contributor Roles Taxonomy]

☐ Conceptualization	☐ Data curation	☐ Formal analysis
☐ Funding acquisition	☐ Investigation	☐ Methodology
☐ Project administration	☐ Resources	☐ Software
☐ Supervision	☐ Validation	☐ Visualization
☐ Writing – original draft	☐ Writing – review & editing	

Reason for this change

2.4 Author change information

Given/first name(s)	
Family/last name	
Email address	
Institution	

Change(s) requested for this author ☐ Remove author ☑ Add new author ☐ Make Corresponding Author

Author's Individual contributions (required for author additions only) see [CRediT Contributor Roles Taxonomy]

☐ Conceptualization	☐ Data curation	☐ Formal analysis
☐ Funding acquisition	☐ Investigation	☐ Methodology
☐ Project administration	☐ Resources	☐ Software
☐ Supervision	☐ Validation	☐ Visualization
☐ Writing – original draft	☐ Writing – review & editing	

Reason for this change

2.5 Author change information

Given/first name(s)	
Family/last name	
Email address	@
Institution	

Change(s) requested for this author ☐ Remove author ☐ Add new author ☐ Make Corresponding Author

Author's Individual contributions (required for author additions only) see [CRediT Contributor Roles Taxonomy]

☐ Conceptualization	☐ Data curation	☐ Formal analysis
☐ Funding acquisition	☐ Investigation	☐ Methodology
☐ Project administration	☐ Resources	☐ Software
☐ Supervision	☐ Validation	☐ Visualization
☐ Writing – original draft	☐ Writing – review & editing	

Reason for this change

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Please enter the full name and details for all the removed authors in the table below, "**Agreement of removed authors**."

In the next table, "**Proposed author list**," enter the full name and details of all the proposed authors *in the order that they should appear* in the publication.

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2) to the addition and/or removal of the authors listed in section 2 and to the revised order of the author list in this section 3, and;

3) that all information provided accurately reflects the authorship of the article.

Agreement of removed authors

Full name	Email address	Signature	Date
Md Salek Miah	saleksta@gmail.com	Signed by: Md Salek Miah 496CE938977147E...	2026-04-05
Maria Bintey Kabir	mbkabir67@gmail.com	Signed by: Maria AB313AEC5E114E3...	2026-04-05
	@		
	@		
	@		

Add additional page(s) if needed.

Proposed author list

Order	Full name	Email address	Signature	Date
01	Md Salek Miah	saleksta@gmail.com	Signed by: Md Salek Miah 4[illegible]6CE9389[illegible]7147E...	2026-04-05
02	Maria Bintey Kabir	mbkabir67@gmail.com	Signed by: Maria AB313AEC5E114E3...	2026-04-05
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Add additional page(s) if needed.

Data Availability

The data used in this study are publicly available from the Demographic and Health Surveys (DHS) Program and can be requested at https://www.dhsprogram.com/data.

Assistance of Generative AI Tools

AI-assisted tools including Grammarly, QuillBot, and ChatGPT were used solely for language editing and grammar improvements. AI did not contribute to data analysis, interpretation, or intellectual content. The authors verified the manuscript for originality and ethical compliance.

CRediT authorship contribution statement

Md Salek Miah: Conceptualization, Data Curation, Methodology, Formal Analysis, Writing – Original Draft and Final Manuscript, Resources, Project Administration.

Funding

This study did not receive any funding from public, private, or non-profit organizations.

Code Availability

All codes and analytical datasets used in this study are available from the first author upon request.

Acknowledgements

We thank the DHS Program for providing free access to the 2022 survey data. We also acknowledge Dr. Jibanul Haque (PhD, University of Central Florida; AI Scientist, The Home Depot) for guidance and support in designing, implementing, and interpreting the machine learning and explainable AI analyses, which improved the rigor and clarity of this study.

Title page (with author information)

Md Salek Miah[a*], Maria Bintey Kabir[b,c]

[a]Department of Statistics, Shahjalal University of Science & Technology (SUST), Sylhet-3114, Bangladesh

[b]Shaheed Suhrawardy Medical College, Dhaka-1207, Bangladesh

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April 5, 2026
To
Co-Editor-in-Chief
Dr. Jair Soares
Journal of Affective Disorders Reports

Subject: Submission of Revised Manuscript

Dear Editor,

Thank you for the opportunity to submit a revised version of our manuscript entitled:
"Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urban Differences."

We sincerely appreciate the constructive and thoughtful comments provided by the reviewers and the editorial team. In response, we have carefully revised the manuscript and prepared both a cleaned and a tracked-changes version to ensure full transparency. Specifically, we have:

- Polished the manuscript for clarity, coherence, and academic rigor.
- Corrected all typographical, spacing, and formatting inconsistencies noted by the reviewers.
- Standardized terminology and improved the presentation of confidence intervals and statistical reporting.
- Revised the Discussion section to enhance interpretive depth, link findings to theory, and align with the study objectives.
- Clarified spatial findings, strengthened conceptual explanations, and improved the logical flow of key sections.
- Updated figures, tables, and references to comply with journal standards.

Additionally, we have included a new co-author to support major revisions and manuscript improvement:
Maria Bintey Kabir

- Shaheed Suhrawardy Medical College, Dhaka-1207, Bangladesh
- Study Physician, Projahnmo Research Foundation, Dhaka-1213, Bangladesh

We are submitting the following files for your consideration:

1. **Miah_Manuscript_2026_Cleaned.docx** – clean revised version
2. **Miah_Manuscript_2026_Revised.docx** – tracked-changes version

3. **Response_Letter.pdf** – point-by-point responses to all reviewer comments with page and line references.

We hope that these revisions adequately address all concerns and further strengthen the clarity and scientific contribution of the manuscript. We are grateful for the reviewers' guidance and look forward to your consideration of the revised version.

Best regards,

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Reviewer 1

Reviewer 1 – Major Concerns and Author Responses

Comment 1:

"What is the depression situation in Lesotho? This is missing in the background section."

Author Response [Page no: 5, Line no: 92-97]

Thank you for this important observation. We have now expanded the background section to clearly describe the burden of depression in Lesotho. Drawing on evidence from DHS Program (2023) and Jejaw et al. (2026), we added that depressive symptoms remain a major public-health concern in Lesotho, with a substantial burden of untreated mental-health conditions among women. Furthermore, based on Myint et al. (2025a), we highlight that Lesotho has one of the highest estimated depressions prevalences in Southern Africa, with women disproportionately affected. These details have now been incorporated into the revised background section.

Comment 2:

"Similarly, what is the state of autonomy in Lesotho? This detail is not provided in the background section."

Author Response [Page no: 5-line no: 100-103]

We appreciate this constructive comment. To address it, we incorporated additional literature that documents the state of women's autonomy in Lesotho. Findings from Gebeyehu et al. (2022), Hormazábal-Salgado et al. (2024), and Viens et al. (2016) have been added to explain that women's autonomy remains notably limited—particularly in decisions related to health care, household purchases, and mobility. These studies also show that rural women experience even lower decision-making power due to entrenched gender norms and economic dependence. This contextual information has been added to strengthen the background section.

Comment 3:

"Why women, not men or both? This justification is needed in the background section."

Author Response [Page 5, Line no:98-99]

Thank you for raising this important point. We have now provided a clearer justification for focusing on women. Drawing on evidence from DHS Program (2023), Myint et al. (2025a), Fernández et al. (2025), and Okyere et al. (2025a), we explain that depression disproportionately affects women in Lesotho, and they exhibit slightly higher help-seeking prevalence (17.1%) compared to men (16.4%). Studies further indicate that women—particularly those in urban areas—are more likely to seek help for depressive symptoms. Combined with limited autonomy and structural gender inequalities, women represent a uniquely vulnerable population requiring focused study. Additionally, autonomy measures in DHS are validated only for women, not men. This rationale has now been clearly integrated into the background section.

Comment 4:

"What is the theoretical basis for this study? Theoretically, what mechanisms does autonomy influence depression?"

Author Response (Page 6, Line 115–121):

Thank you for this insightful comment. We have now added a brief theoretical explanation grounded in **Self-Determination Theory** and **Gender and Power Theory**. These frameworks explain that limited autonomy restricts women's ability to make decisions about their health, mobility, and resources, which increases psychological stress, reduces coping capacity, and heightens vulnerability to depressive symptoms. We have incorporated these mechanisms into the background section to clarify the theoretical basis of our study. The theoretical linkage has now been added to the revised manuscript.

Reviewer Comment 5 :

"What sensitivity analysis was performed given that the PHQ-9 was used?"

Author Response (Page 12, Line 254–255):

Thank you for raising this important point. In the current study, we did not perform additional sensitivity analyses specifically for the PHQ-9 scoring. This is because we followed the standard DHS scoring protocol, which applies the validated PHQ-9 cut-off for moderate-to-severe depressive symptoms. However, we acknowledge that alternative cut-offs or item-level adjustments could provide additional robustness. We have now added a clarification in the Methods section to explain this and noted the lack of sensitivity analysis as a limitation. We appreciate the reviewer's insight, which will inform future work.

Reviewer Comment 6 :

"The description of the exposure needs much clarification."

Author Response (Page 8–9, Line 160–176):

We thank the reviewer for this important comment. We have clarified the measurement and categorization of decision-making autonomy to ensure transparency and reproducibility. The exposure variable is now described in detail, including its domains, scoring procedures, and the steps used to derive the final binary variable for analysis.

Reviewer Comment 7:

"What influenced the choice of covariates? This is missing in the methods section."

Author Response (Page 9–10, Line 179–203):

We thank the reviewer for highlighting this important point. The covariates included in our study were selected based on prior evidence and standard DHS analytical practices, reflecting sociodemographic, reproductive, household, and socioeconomic factors known to influence women's mental health outcomes (Esteban-Gonzalo et al., 2026; M. S. Miah et al., 2026; Tamanna et al., 2025; Zoha et al., 2025). These variables were harmonized with previous DHS-based studies to ensure comparability across analyses and to appropriately control for potential confounding when examining the association between decision-making autonomy and depressive symptoms.

Reviewer Comment 8:

"Line 249: Clarify this confidence interval (AOR = 0.44; 95% CI: 0.271974)"

Author Response (Page 15, Line 324):

We thank the reviewer for pointing this out. We have corrected the formatting of the confidence interval for improved clarity. The revised text now reads: AOR = 0.44 (95% CI: 0.27–0.97), ensuring accurate and reader-friendly interpretation of the reported association.

Reviewer Comment 9:

"Lines 275–279: There is no need repeating the results in the discussion section."

Author Response (Page 17, Lines 353–358):

We thank the reviewer for this insightful comment. We have revised the discussion section to minimize redundancy and have removed sentences that repeated descriptive results. The discussion now focuses on interpreting the findings, linking them to relevant theoretical frameworks, and emphasizing their implications rather than restating earlier results.

Reviewer Comment 10:

"The discussion section should be anchored in theory."

Author Response (Page 20, Lines 426–430; Page 19, Lines 395–399):

We thank the reviewer for this valuable comment. In response, we have revised the discussion section to more clearly anchor our interpretation of findings within established theoretical frameworks. Specifically, we have incorporated the Social Determinants of Mental Health framework to explain how structural inequities, gendered power dynamics, and levels of decision-making autonomy influence depressive symptoms among women. These theoretical linkages strengthen the interpretability of the results and provide a more coherent explanation of the observed rural–urban differences.

Reviewer Comment 11:

"What accounted for the spatial variations? This is a key finding of your study and so it needs to be properly discussed."

Author Response (Page 20, Lines 426–430):

We appreciate the reviewer's insightful observation. In response, we have expanded the discussion to more fully explain the underlying factors contributing to the observed spatial variations. Specifically, we discuss how geographic disparities reflect differences in women's decision-making autonomy, socioeconomic inequalities, rural–urban structural conditions, and district-level contextual stressors. These contextual factors help to clarify why certain provinces exhibit higher depressive symptom burdens than others and strengthen the interpretation of our spatial findings.

Reviewer Comment 12:

"Discuss the limitations of using the PHQ-9 in such a study."

Author Response (Page 21-22, Lines 455–458):

We thank the reviewer for highlighting this important point. We have now added a clear discussion of the limitations associated with the use of the PHQ-9 in this context. Specifically, we note that the PHQ-9 is a self-reported screening tool that may be subject to recall and reporting biases, and that its sensitivity and optimal cut-off points can vary across cultural and population settings. Nevertheless, we emphasize that the PHQ-9 remains one of the most widely validated and commonly used instruments for assessing depressive symptoms in large population-based surveys. These considerations have been incorporated into the revised manuscript.

Reviewer Comment 13:

"The presentation of the discussion is generally problematic. It should address what was known, what this study adds and the implications thereof."

Author Response (Page 20, Lines 437–439; Page 17–18, Lines 364–382):

We thank the reviewer for this valuable and constructive comment. We have substantially revised the discussion section to improve clarity, structure, and coherence. The revised version now explicitly follows a structured narrative that:

(i) Summarizes what was previously known regarding depressive symptoms and women's decision-making autonomy in LMICs, drawing on existing empirical and theoretical literature;

(ii) Clearly articulates the new contributions of the present study, including the protective role of autonomy, rural–urban heterogeneity, and spatial disparities unique to the Lesotho context; and

(iii) Outlines the implications of these findings, emphasizing the need for targeted, context-specific mental health and empowerment interventions informed by spatial and sociodemographic patterns.

These revisions make the discussion more concise, theoretically grounded, and aligned with the journal's expectations for clear interpretation and contribution to the field.

Closing Statement

We sincerely thank **Reviewer 1** for the thoughtful and constructive feedback, which has significantly improved the clarity, rigor, and interpretability of our manuscript. We believe that all concerns have been carefully addressed, and the revisions have strengthened the study's contribution to understanding the role of women's decision-making autonomy and depressive symptoms in Lesotho. We greatly appreciate your time and insightful comments.

Reviewer 2

Reviewer Comment 1:

Confidence interval formatting – Several confidence intervals appear without appropriate spacing or punctuation, which may confuse readers. For example: "7.40% (95% CI: 6.268.74)" and "51.4758.45". Please revise to standard formats such as "6.26-8.74" or "51.47-58.45".

Response to Reviewer 2:

We sincerely thank the reviewer for this careful observation. We have thoroughly reviewed all confidence intervals throughout the manuscript and corrected their formatting to follow standard conventions (e.g., "6.26–8.74" and "51.47–58.45") to ensure clarity, consistency, and accurate interpretation [***Page 13, Line 272, 273, 276, 278***].

Reviewer Comment 2:

Typographical and spacing inconsistencies A few lines contain missing parentheses or spacing errors, particularly in the Results and Table sections. Example: "0.83 0.56-1.23) 0.368". A consistent format such as "0.83 (0.56-1.23), p = 0.368" would improve clarity.

Response to Reviewer 2:

We sincerely thank the reviewer for this careful observation. All typographical and spacing inconsistencies, including parentheses and p-value formatting in the Results and Table 2, have been thoroughly corrected to ensure uniformity, clarity, and improved readability [Page 42, Line 823–832, **Table 2**].

Reviewer Comment 3:

Consistency in terminology for depressive symptoms – The manuscript alternates between "moderate-severe," "moderate-to-severe," and "moderate to severe." For example: "A PHQ-9 *score of 10 and above was considered a moderate-severe depressive symptom." Please standardize terminology throughout*.

Response to Reviewer 2:

We thank the reviewer for this helpful comment. We have carefully standardized the terminology throughout the manuscript to use **"moderate-to-severe" depressive symptoms** consistently, ensuring clarity and uniformity across all sections.

Reviewer Comment 4:

Clarification of spatial findings wording – The description of autonomy patterns includes slightly contradictory phrasing: "lower levels of autonomy were centralized towards the southern and south-western provinces… but autonomy was nevertheless poor in general." Consider clarifying whether autonomy is uniformly low or varies meaningfully across regions.

Response to Reviewer 2:

We sincerely thank the reviewer for this valuable suggestion. We have revised the text for clarity as follows:

Updated text:

"Lower levels of autonomy were concentrated in the southern and south-western provinces (Quthing, Qacha's Nek, Mohale's Hoek, and Mafeteng), which represent high-burden areas of limited decision-making power. In contrast, the central and northern provinces, including Thaba-Tseka and Maseru, exhibited relatively higher autonomy, although overall empowerment levels remained modest. These geographic patterns highlight meaningful regional variation in female autonomy and identify priority areas for context-specific interventions."

This revision clarifies that autonomy varies across regions while emphasizing that overall empowerment is still limited [**Page 16, Line 331–336**].

Reviewer 2 – Comment 5:

Minor improvements to flow and readability

A few sentences in the Results section are long and could be streamlined for clarity. For example, the paragraph describing age patterns, reproductive stressors, and socioeconomic variation could be broken into shorter, more focused sentences to enhance readability.

Response to Reviewer 2:

We thank the reviewer for this helpful suggestion. We have revised the Results section to improve clarity and readability by breaking long sentences into shorter, more focused statements, particularly in the paragraphs describing age patterns, reproductive stressors, and socioeconomic variations. These edits enhance the flow and make the findings easier to follow without altering their meaning [Page 14, Line 283-292].

Closing Statement

We sincerely thank the reviewer for their careful and constructive feedback. All suggested revisions, including formatting, typographical corrections, clarity in spatial findings, and improvements in discussion and statistical reporting, have been carefully addressed. We believe these changes have significantly enhanced the clarity, readability, and scientific rigor of the manuscript. We greatly appreciate the reviewer's guidance and thoughtful comments, which have helped improve the overall quality of our work.

Declarations

Consent to participant

Not applicable.

Consent for Publication

Not applicable. This study used de-identified, publicly available data and does not include any identifiable information.

Conflict of Interest

The authors declare no competing interests.

CRediT authorship contribution statement

Md Salek Miah: Conceptualization, Data Curation, Methodology, Formal Analysis, Writing – Original Draft and Final Manuscript, Resources, Project Administration.

Funding

This study did not receive any funding from public, private, or non-profit organizations.

Code Availability

All codes and analytical datasets used in this study are available from the first author upon request.

Acknowledgements

We thank the DHS Program for providing free access to the 2022 survey data. We also acknowledge Dr. Jibanul Haque (PhD, University of Central Florida; AI Scientist, The Home Depot) for guidance and support in designing, implementing, and interpreting the machine learning and explainable AI analyses, which improved the rigor and clarity of this study.

Highlights

- Depression affected 7.4% of women, with marked rural–urban and geographic inequalities.
- Higher decision-making autonomy was associated with lower odds of depression, particularly in rural areas.
- Spatial clustering revealed high-burden districts, indicating the need for targeted, context-specific interventions.

Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urban Differences

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Abstract

Objectives:

To assess the association between depressive symptoms and decision-making autonomy with rural urban inequalities, and mapped province-level spatial disparities in depression prevalence in ever-married women in Lesotho.

Methods:

This study used cross-section data of 3,297 women aged 15–49 years from the 2023–24 Lesotho Demographic and Health Survey (LDHS). Depression was measured using the Patient Health Questionnaire-9 (PHQ-9 ≥10). Women's decision-making autonomy was measured using a composite empowerment index based on participation in five household decision-making domains. Survey-weighted multivariable logistic regression models were fitted at the national level and stratified by place of residence. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC). Spatial analyses were conducted to examine district-level variation in depression prevalence and autonomy.

Results:

Overall, 7.4% (95% CI: 6.3–8.7) of women reported moderate-severe depressive symptoms. Women with decision-making autonomy showed lower odds of depression nationally (AOR = 0.63; 95% CI: 0.39–0.98), with a stronger effect in rural areas (AOR = 0.44; 95% CI: 0.27–0.74). Depression prevalence was highest in central–northern districts (Maseru, Berea, Leribe, Butha-Buthe) and lower levels of autonomy observed in southern and south-western provinces.

Conclusions:

Women's decision-making autonomy is significantly associated with lower odds of depressive symptoms in Lesotho, particularly among rural women. The observed rural–urban and spatial inequalities highlight the need for context-specific mental health and reproductive health interventions that incorporate women's empowerment as a core component.

Keywords: Depression, Decision-making autonomy, Pregnancy loss, Rural–urban disparities, Lesotho, Spatial analysis

Highlights

- Depression affected 7.4% of women, with marked rural–urban and geographic inequalities.
- Higher decision-making autonomy was associated with lower odds of depression, particularly in rural areas.
- Spatial clustering revealed high-burden districts, indicating the need for targeted, context-specific interventions.

Introduction

Depression is one of the causes of disability in the whole world and it is even higher among women in reproductive age (Herrman et al., 2022). Depression is more common and prevalent in women than men because of an aggregate of biological, social, and environmental issues (Patel et al., 2018a). Social roles are gendered, and access to economic resources is limited in low-resource settings, making people more vulnerable to mental health problems due to reproductive roles and restricted access to mental health (Patel et al., 2018b).

Women possessing depression do not only impact the quality of life, but the functioning of the family, child health, and well-being of the community, in general (Oram et al., 2022). It is thus important to understand the social determinants that affect the depressive symptoms in order to bring meaningful, context-sensitive intervention (Amouzou et al., 2025). The decision-making autonomy of women is one of the important social determinants of mental health (The Lancet, 2025).

Decision-making autonomy is defined as a woman having power to engage in making decisions regarding her health care, family purchases, finances, mobility and reproductive decisions (Heise et al., 2019). The results of numerous nations show that women who have more freedom are less likely to encounter depressive and anxiety symptoms (Westby et al., 2021). Autonomy can keep mental health safe because it would enhance the sense of control that women have in their lives, decrease their vulnerability to coercive relationships, and allow access to health and social resources (Leight et al., 2022a). On the other hand, a lack of autonomy has been associated with higher psychological distress, helplessness and increased vulnerability to common mental disorders (Leight et al., 2022b; M. S. U. M. O. Miah, 2025). Although the increased awareness of the need of autonomy among women and the mental wellness of women has advanced throughout the globe, few studies have been conducted in most low-income countries (LMICs) (Borges do Nascimento et al., 2025; Jia et al., 2025). Mental health of women in sub-Saharan Africa is a product of socioeconomic, reproductive, and contextual risk factors (Bandara et al., 2024). There are early marriage, early childbearing, internet use, pregnancy loss, high fertility and large family size, which are usually stressful factors that can make an individual vulnerable to depression (M. S. Miah & Ullah, 2025). These risk factors are usually accompanied by limited decision-making autonomy especially in rural regions where traditions and gender norms are the order of the day

(Rahman et al., 2024). Although urban setting usually has more access to education, work, and health services, other stressors, such as economic insecurity, social isolation, and accelerated sociocultural change, might also be brought about by urban setting (Nabaggala et al., 2021). These aspects indicate that the interdependence between autonomy and depression might depend on the setting and there is a necessity to explore the rural-urban variations in mental health outcomes (Acharya et al., 2010b, 2010a; Idris et al., 2023; Mustafiz et al., 2025).

In Southern Africa, in Lesotho, depressive symptoms remain a major public health issue, with recent national surveys reporting a substantial burden of untreated mental health conditions among women, driven by digital access, poverty, human immunodeficiency virus (HIV) prevalence, and limited access to mental-health services (DHS Program, 2023; Jejaw et al., 2026). Additionally, Lesotho showed the highest estimated prevalence of depression, with women excessively suffered (Myint et al., 2025a). Likely, a slightly higher prevalence of women sought (17.1%) help for depression compared to men (16.4%). Particularly, women's living in urban's settings showed higher help-seeking prevalences compared to men (Fernández et al., 2025; Okyere et al., 2025a). Previous studies also shows that women's autonomy in Lesotho remains remarkably inadequate, mainly in decisions related to health care, household purchases, and mobility, with rural women experiencing even lower levels of intervention due to entrenched gender norms and economic dependence (Gebeyehu et al., 2022; Hormazábal-Salgado et al., 2024; Viens et al., 2016). In Lesotho, limited studies exist about the mental health of women and their social determinants (DHS Program, 2023). There has been a preponderance of studies which have concentrated on maternal and reproductive health outcomes and little has been done on psychological well-being (Leseba et al., 2025). Naturally, there is scanty evidence on the association between decision-making autonomy with depressive symptoms among women in national representative samples

(Mokwena & Makhozonke, 2024). In addition, the rural-urban disparities or province-wise spatial inequalities in the prevalence of depression have not been investigated in previous studies, which are important in determining the areas of high burden and effective intervention strategies (Okyere et al., 2025b). This gap in evidence is a major shortcoming since there is a need to combine policy and programmatic planning in Lesotho to address the issues of women empowerment and mental health (Myint et al., 2025b).

Theoretically, women's autonomy influences mental-health outcomes through multiple mechanisms. Guided by Self-Determination Theory and Gender and Power Theory, limited autonomy reduces women's capacity to make independent decisions regarding health care, household resources, and mobility. Such constraints increase psychological stress, diminish perceived control, weaken coping ability, and ultimately elevate the risk of depressive symptoms (Hindman et al., 2026; Raz et al., 2025). These theoretical perspectives support the link between autonomy and depression in the context of Lesotho (DHS Program, 2023).Against this background, the present study aimed to examine the association between decision making autonomy and moderate-severe depressive symptoms among women in Lesotho using nationally representative LDHS data (DHS Program, 2023). Specifically, this study assessed this association at the national level, conducted stratified analyses by rural and urban residence to explore inequalities, and mapped province-wise spatial variation in depression prevalence. By integrating individual-level regression analyses with spatial methods, this study seeks to provide policy-relevant evidence to inform targeted mental health and reproductive health interventions in Lesotho.

Methodology

Setting, study design and source of data

This paper is based on a secondary analysis of the data of the 2023-24 Lesotho Demographic and Health Survey (LDHS), a household survey that was conducted nationally by the Lesotho Ministry of Health with technical assistance of ICF under the DHS Program. The data was collected from 27 November 2023 to 29 February 2024. To present current estimates of the demographic and health variables, such as fertility, maternal and child health, and mental health according to the Sustainable Development Goals (SDGs). The LDHS utilized standardized DHS questionnaires (household, women, men and biomarker) that were translated to the Lesotho setting and were conducted in the Sesotho and English languages. Face to face interviews were conducted using tablet-based computer aided personal interviewing (CAPI) and inbuilt quality checks were used (DHS Program, 2023).This study followed **S**trengthening the **R**eporting of **O**bservational **S**tudies in **E**pidemiology (STROBE) guidelines (Cuschieri, 2019).

Sampling design and population of study

LDHS used a stratified cluster sampling design, which was two-stage using the 2016 Lesotho Population and Housing Census. At the first level, 400 enumeration areas (EAs) were chosen on a basis of probability proportional to size. Twenty-five households were sampled per EA using systematic random sampling as per the second stage. All eligible persons were all women aged 15 49 years, and were usual residents of the selected households, or had stayed in the selected households the night before the survey. This was an analysis of women who were on subsampled households and who had completed PHQ-9 depression module. The weighted analytic sample was the final sample of 3, 297 ever-married women aged 15- 49 [**Figure 1**].

[**Insert Figure 1 Here**]

[Figure 1: The Study Flow]

Outcome variable

The main outcome variable Patient Health Questionnaire-9 (PHQ-9) was used to evaluate the depressive symptoms. A PHQ-9 score of 10 and above was considered a moderate-severe depressive symptom in women whereas a score below 10 was considered no-depression. All regression analysis used this binary outcome. Internal consistency of the PHQ-9 in this sample was assessed using Cronbach's alpha ($\alpha = 0.82$), indicating good reliability. (M. S. Miah & Ullah, 2025; Westby et al., 2021).

Exposure Variable Decision-Making Autonomy

The main exposure variable was decision-making autonomy which was measured by a composite index of empowerment based on five domains of household and reproductive choices:

1. Own health care.

2. Large household purchases.

3. Use of household money.

4. Visits to relatives or family.

5. Family planning decisions.

Women were regarded autonomous in each domain if they reported making the decision alone or jointly with their partner. Each empowered domain contributed 1 point, resulting in a composite score from 0 to 5. Then, this study considered, no autonomy: composite score ≤ 2 and yes autonomy: composite score ≥ 3. This binary variable (Decision-Making Autonomy) was used as the main exposure in all analyses examining its association with depressive symptoms. This method corresponds with the past research conducted by the DHS on the level of empowerment of women and represents the meaningful involvement in the household decision-making process. his binary measure captures the overall level of meaningful involvement in household and reproductive decisions, providing a clear indicator of women's autonomy for analysis (Ameyaw et al., 2016; Anfaara et al., 2024; Antabe et al., 2025; Kamiya, 2011; M. S. U. M. O. Miah, 2025).

Covariates

In this study, women's sociodemographic characteristics were categorized in line with standard DHS methods. Women's age was categorized into seven groups (15–19, 20–24, 25–29, 30–34, 35–39, 40–44, 45–49 years), while age at first birth was grouped as early (10–17 years), normal (18–24 years), and late (≥25 years). Age at first sexual intercourse was categorized as <18 years or ≥18 years, and age at cohabitation was classified into young (<18 years), mild (18–20 years), and older (≥21 years). Women's education was categorized into below primary, secondary incomplete, and secondary or higher, and maternal employment was categorized as currently working or not working. Adequate antenatal care (ANC) visits were dichotomized as <4 or ≥4 visits. Reproductive characteristics included number of children (none, 1–2, 3+), birth size (small, average or large), preterm birth (term ≥9 months, preterm <9 months), current pregnancy status (yes/no), pregnancy multiplicity (single/multiple), pressure to be pregnant (yes/no), and history of pregnancy loss (none, one loss, two or more losses). Residing status was classified as residing with or not with husband/partner, current abstinence as yes/no, number of unions as once or more than once, and recent menstruation as yes/no. IPV justification (does not justify/justifies IPV). Household environment and socioeconomic factors included place of residence (urban/rural), administrative province (Butha-Buthe, Leribe, Berea, Maseru, Mafeteng, Mohale's Hoek, Quthing, Qacha's Nek, Mokhotlong, Thaba-Tseka), religion (Roman Catholic/Other), wealth status (poor, middle, rich), household size (1–3, 4+ members), household head sex (male/female), household assets (ownership of motorbike or bicycle; yes/no), media access (access to TV, radio, newspaper, or mobile; yes/no), and household construction materials (based on roof materials; improved/unimproved). Source of drinking water and sanitation facility were also categorized as improved or unimproved. Internet use in the last 12 months was coded as yes/no. All covariate categories were harmonized with previous DHS-based studies to ensure consistency in reporting and comparability across surveys and analyses (Esteban-Gonzalo et al., 2026; M. S. Miah et al., 2026; Tamanna et al., 2025; Tamanna & Mahmud, 2025; Zoha et al., 2025).

Data quality and management

The data were entered electronically (CSPro) and checked automatically in terms of range and consistency. Supervision of data on a daily basis, transfer in safe data and central data cleaning guaranteed high data quality. Additional checking and cleaning of secondary data were performed before final analysis.

Statistical analysis

All the analyses factored in the complicated survey design with sampling weights, clustering, and stratification in order to make it representative of the nation. An analysis of complete-cases was applied.

Code Availability

The cleaned dataset, analysis scripts, and related resources supporting the findings of this study are available at GitHub:

https://github.com/muhammadsalek/Rural-Urban-Inequalities-in-Depression-among-Ever-Married-Women-in-Lesotho. The analyses were conducted following standard public health and epidemiological approaches and are consistent with methods applied in previous literature (Celentano ScD MHS & Szklo MPH DrPH, 2019).

Descriptive, and bivariate analysis

The characteristics of the participants were summarized using descriptive statistics. Chi-square tests were used to test bivariate relationship between depressive symptoms and explanatory variables. Multivariate modeling was considered on variables having p less than 0.20 or good conceptual relevance (Celentano ScD MHS & Szklo MPH DrPH, 2019). Spatial analysis to depict spatial difference in the prevalence of depression and autonomy of women in decision-making at the province level, choropleth maps were shaped.

Multivariable modeling

Logistic regression was used to assess associations between depressive symptoms and decision-making autonomy. Models were built sequentially using a forward stepwise approach, informed by both statistical significance and prior literature:

- **Model 0:** Maternal age and decision-making autonomy variable just.
- **Model 1:** Sociodemographic factors (education, religion, residence).
- **Model 2:** Reproductive factors (age at first birth, cohabitation, parity, multiple pregnancy, pregnancy loss).
- **Model 3:** Contextual factors (urban/rural residence)
- **Model 4 (Final model):** Household and environmental factors (toilet facility, water source, household size, household head sex, wealth, media and internet access, household assets and materials).

The final model was nominated based on conceptual relevance and improved model fit (AIC = 1648.22; BIC = 1886.15; mean VIF = 2.2).

Final Model Adjusted Covariates

This set of confounders were adjusted in the final balanced model including women age, women education, maternal employment, religion, administrative division, age at first birth, age at cohabitation, number of children, pregnancy status, pregnancy loss, place of residence, toilet facilities, water access, household head sex, household size category, household wealth status, media access, internet use, household assets, and household materials.

Model diagnostics

Multicollinearity was assessed using variance inflation factors (VIF). Model fit was assessed using the Hosmer–Lemeshow test, which showed good fit ($\chi^2 = 5.85$, $p = 0.6642$). Model discrimination was assessed using the area under the ROC curve (AUC = 0.6781), demonstrating moderate predictive ability. Results are reported as adjusted odds ratios (aORs) with 95% confidence intervals, with statistical significance set at $p < 0.05$.

Sensitivity Analysis

In this study, no additional sensitivity analyses were conducted for the PHQ-9 scores, as we adhered to the standard DHS scoring protocol for defining moderate-severe depressive symptoms (DHS Program, 2023).

Ethical considerations

The LDHS protocol acknowledged ethical approval from relevant ethics committees in Lesotho and the ICF Institutional Review Board. Written informed consent was obtained from all participants. This study used anonymized, publicly available DHS data with permission from the

DHS Program. However, details available at at https://dhsprogram.com/Methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm .

Results

The prevalence of depression was 7.4% (95% CI: 6.3-8.7). On place of residence, just over half (54.98%) respondents reported the rural (95% CI: 51.47-58.45) and 45.02% reported the urban area (95% CI: 41.55-48.53). The small overlap between the confidence limits implies that more rural participants are represented in the study population though, to establish the statistical significance of the difference, this would have to be tested. The confidence intervals of the two categories of depression and non-depression are complementary as expected, and it proves that depression is a minor yet non-trivial burden to the population health [**Figure 2,Figure 3**].

[**Insert Figure 2 and Figure 3 Here**]

[Figure 2 : Prevalences of Depression Symptoms]

[Figure 3: Rural-Urban Inequalities of Depression Prevalence]

The prevalence of the rural population in the study population was 54.98% (95% CI: 51.47-58.45) and that of the urban population was 45.02% (95% CI: 41.55-48.53). The fact that the CIs are mainly non-overlapping implies that the percentage of rural members might be higher than that of urban people, yet it is necessary to test whether the difference is significant. The depression prevalence was 7.40% (95% CI: 6.26-8.74) meaning it is possible to suggest that depression can be experienced by 6-9% of the target population. On the other hand, 92.60% respondents were not depressed (95%CI: 91.26-93.74), which is an accurate estimate since the percentage in this group was large. All in all, these results suggest that the prevalence of depression is lower than in the majority of the population, and rural inhabitants are a slightly higher proportion of the study sample [**Figure 4**].

[**Insert Figure 4 Here**]

[Figure 4: Rural-Urban Inequalities of Decision -Making Autonomy]

On the whole, depressive symptoms were observed only in a few prevalences of women. Particularly, higher among women in the age groups of 15-19 years, 25-29 years, and 35-39 years. On the other hand, early reproductive such as early fertility, early onset of sexual activity, and early cohabitation were associated with relatively high levels of depressive symptoms. The variations due to socioeconomic status were relatively low, with women having lower education levels and being unemployed displaying relatively high prevalence. In relation to health determinants, limited variation was noted apart from women who had relatively fewer antenatal visits and women with high parity having a higher prevalence of depressive symptoms. Women who lacked decision-making power and those from urban centers had relatively high depression levels. The variations by district were also significant **[Table 3**].

[**Insert Table 1 Here**]

[**Table 1: Distribution of depressive symptoms (PHQ-9 ≥10) by sociodemographic and reproductive characteristics (N=3,297)]**

When controlling the odds of depression against a comprehensive set of sociodemographic, reproductive, household, and contextual confounders, women with an autonomy of decision made were revealed to have much lower odds of depression at the national level (AOR = 0.63; 95% CI: 0.39-0.98, p=0.048). Additionally, there was a significant rural-urban heterogeneity indicated by stratified analyses. Decision-making autonomy in rural areas was significantly associated to lower chances of having depressive symptoms (AOR = 0.44; 95% CI: 0.27-0.74, p=0.002) only [**Table 4**].

[**Insert Table 2 Here**]

[**Table 2:Associations Between Decision Making Autonomy and Depression Symptoms (N= 3,297, N1 Rural= 2,086, N2 Urban =1,211)**]

An AUC of 0.68 implies that, in approximately 68% of randomly selected case–non-case pairs, the model assigns a higher predicted probability to the woman with the outcome than to the woman without the outcome [**Figure 5**].

[**Insert Figure 5 Here**]

[Figure 5:ROC curve showing model performance]

Lower levels of autonomy were concentrated in the southern and south-western provinces (Quthing, Qacha's Nek, Mohale's Hoek, and Mafeteng), which represent high-burden areas of limited decision-making power. In contrast, the central and northern provinces, including Thaba-Tseka and Maseru, exhibited relatively higher autonomy, although overall empowerment levels remained modest. These geographic patterns highlight meaningful regional variation in female autonomy and identify priority areas for context-specific interventions [**Figure 7**].

[**Insert Figure 7 Here**]

[Figure 6: Spatial Inequalities of Depression Prevalences]

The central-northern districts, especially the Maseru, Berea, Leribe, and Butha-Buthe, proved to have a higher prevalence of depression, which resulted in the identification of these districts as high-burden areas. Conversely, it was showed to be lower in the southern and south-eastern districts, which comprise of Quthing, Qachas Nek, Mafeteng, and some parts of Hoek of Mohale which are relatively low-burden areas. These geographic inequalities reveal that there exist significant subnational differences in mental health burden and that mental health planning and resource allocation needs to be done at the district level with a focus on more burdensome central-northern districts [**Figure 6**].

[**Insert Figure 6 Here**]

[**Figure 6: Spatial Inequalities of Decision-making Autonomy]**

Discussion

Using nationally representative LDHS 2023–24 data, this study found that women in Lesotho experienced moderate-severe depressive symptoms, confirming a non-trivial mental health burden. Spatial analysis revealed substantial geographic inequalities: central–northern districts, including Maseru, Berea, Leribe, and Butha-Buthe, had the highest prevalence of depression, whereas southern and south-western districts (Quthing, Qacha's Nek, Mohale's Hoek, Mafeteng) exhibited the lowest autonomy and comparatively lower depression prevalence. These findings highlight the importance of women's decision-making autonomy for mental health, particularly in rural areas, and underscore spatial disparities that warrant targeted interventions. These findings are interpreted within the social determinants of mental health framework, highlighting those structural inequities, reproductive stressors, and decision-making autonomy shape women's vulnerability to depressive symptoms in Lesotho.

Previous studies have shown that depressive symptoms are a growing concern in Southern Africa, particularly among women facing social and economic vulnerabilities (Melkam et al., 2025; Mokwena & Makhozonke, 2024). Our study adds new evidence by linking decision-making autonomy directly to depressive symptoms at both national and spatial levels, highlighting rural–urban and geographic disparities (Noor, Siddique, Yeasar, et al., 2025).According to our findings, the sample proportion of rural residents was a little bit higher (55%), but the rates of depressive symptoms were slightly higher in women in the urban group. The observed rural–urban differences may reflect the distinct social and economic environments in Lesotho. Although rural women are not easily accessible to professional mental health care providers, their social and family networks provide them with more advantages in terms of making decisions and dealing with problems, increasing the protective value of autonomy (Acharya et al., 2010b; Ameyaw et al., 2016). On the other hand, even though urban women have more resources at their disposal, they are more prone to economic hardship, social isolation, and urban stress, all of which reduce the protective nature of autonomy on depression (Antabe et al., 2025). This is contrary to other LMIC studies where urban women can usually report improved mental health outcomes because of increased socioeconomic resources (Asmare et al., 2026). However, in Lesotho, urban life can be accompanied by a greater exposure to stressors, including economic, social isolation or urban poverty, which may at least partly be responsible for the observed urban vulnerability (Mokwena & Makhozonke, 2024). It was determined that events and pressures in early life including the earlier age of first birth, earlier sexual initiation, earlier cohabitation, and pregnancy loss were connected with increased depressive symptomatology (Moeti et al., 2022; Russell et al., 2021). These results are in line with the previous studies that associate reproductive stressors and early life transitions to mental health vulnerability in the long term as a woman (Bushu et al., 2025). It is important to note that the association between the depressive symptoms and pregnancy loss highlights the urgent necessity of providing women with reproductive difficulties with a specific kind of psychosocial assistance (Boakye et al., 2025).

Taken together, these findings suggest that empowering women in decision-making and addressing early-life reproductive stressors are critical for reducing depressive symptoms. The implications for policy include designing context-specific, equity-oriented interventions that combine reproductive and mental health support, with particular attention to rural areas and high-burden districts.

Less education and unemployment and poor household wealth were also associated with it but with slightly high prevalence of depression, though it was within the same confidence interval (Quenby et al., 2021). This is in line with the current social determinants of mental health framework that identifies structural inequities as a way of aggravating the risk factors on the individual level (Alkema et al., 2016). In the national level, autonomy in making decisions showed lower odds with depression, and this association was even stronger in rural settings. The results are consistent with other studies on Southern Africa that showed women empowerment and autonomy to be protective against depressive symptoms (Miller et al., 2016). The predictive performance of the multivariable model was modest around (AUC = 0.68), indicating fair discrimination. While the model performs better than chance, it does not achieve a level of accuracy sufficient for individual-level prediction. This is consistent with prior population-based mental health studies, where complex psychosocial outcomes are influenced by unmeasured factors such as interpersonal relationships, trauma, and social norms (Ostovar et al., 2025). As another finding of our spatial analyses, central-northern districts showed more depression prevalence whereas southern and south-western provinces (Quthing, Qacha Nek, Mafeteng, Mohale Hoek) showed more autonomy and less depression (DHS Program, 2023; Mokwena & Makhozonke, 2024). These trends can be aligned with the hypothesis that geographic disparities in enabling and psychosocial stress are associated with unequal mental health workloads in districts (Thibaut & ELNahas, 2023). Interestingly, health-related variables like antenatal care, parity and multiple pregnancies were found to have minimum dependence on depression whereas media and internet exposure did not seem to have any effects on the symptom patterns (Flor et al., 2026). This is contrary to other LMICs where utilization of health services tends to mediate the mental health outcomes implying that the context of social and other determinants of mental health may play a higher role in the Lesotho context(M. S. Miah & Ullah, 2025; Noor, Siddique, Das, et al., 2025; Pandey et al., 2024). Altogether, these results demonstrate the need of context-specific, equity-oriented interventions, which combine both reproductive and mental health support (Zhang et al., 2019). The observed spatial variations, with higher depression prevalence in central–northern districts and lower prevalence in southern and south-western provinces, are likely due to geographic differences in women's autonomy, socioeconomic resources, psychosocial stressors, and urban-rural contextual factors. These findings revealed the need for geographically targeted mental health interventions (Abedin & Arunachalam, 2020). The identified geographic and sociodemographic differences also suggest that national mental health policy must focus on central-northern districts and women with early-life reproductive stressors, as well as encourage autonomy in decision-making as a protective factor (Llop-Gironés et al., 2019). The combination of spatial and individual-level analysis in this research offers practical piece of evidence in order to guide the allocation of resources and specific mental health programming in Lesotho (Dangare et al., 2026).

Overall, this study demonstrates what was previously known, adds novel insights into the protective role of autonomy and geographic disparities, and underscores the need for tailored, empowerment-focused mental health interventions in Lesotho.

Policy Implication and Recommendations

The findings indicate that mental health services in Lesotho need to be provided in context-specific manner. Policies should: Empower rural women who have a low autonomy, improve empowerment initiatives, reproductive decision-making support, and intervention aimed at psychosocial. Increase central-northern districts with high burden of mental health in need of expanded services, community awareness, and primary care integration. Integrate the empowerment strategies into the wider maternal and reproductive health programs, recognizing the differing impacts between the rural and the urban areas.

Strengths and Limitations

The strengths are the utilization of national representative data, the measurement of depression based on PHQ-9, and the incorporation of the spatial analysis that can be used to demonstrate geographic inequities. This study has several limitations, such as its cross-sectional design, which excludes the possibility of causal inference, and use of self-reported measures of autonomy and depression, which is prone to reporting bias. In spite of these shortcomings, the article presents strong evidence on how the issues of empowerment, reproductive factors and mental health intersect in Lesotho. Additionally, the PHQ-9 relies on self-reported symptoms and may not fully capture clinical depression, and its sensitivity and specificity can vary across cultural and demographic contexts. Future studies may consider alternative PHQ-9 cut-offs or item-level sensitivity checks to further assess robustness.

Conclusions

Despite these limitations, the study revealed strong evidence on the association between decision making autonomy and depressive symptoms with rural-urban disparities in Lesotho. These results indicate that the autonomy of women can be a protective factor against depression in rural areas, where the structural and social limitations to the agency of women may be higher. On the whole, the findings reveal context-specific mental health gains of women empowerment and the need to incorporate in rural maternal and mental health interventions autonomy-enhancing strategies. The results indicate that there exist high rural-urban inequalities, geographic inequalities, central-northern districts had prevalence of depression and lowest autonomy was observed in southern-southwestern districts. These findings highlight the importance of context-sensitive, empowerment-based mental health services, especially the rural women and high-burden districts.

The combination of women empowerment strategies with reproductive health and mental health initiatives might help to decrease the symptoms of depression and help to ensure equality in mental health outcomes nationwide.

Declarations

Data Availability

The data used in this study are publicly available from the Demographic and Health Surveys (DHS) Program and can be requested at https://www.dhsprogram.com/data.

Assistance of Generative AI Tools

AI-assisted tools including Grammarly, QuillBot, and ChatGPT were used solely for language editing and grammar improvements. AI did not contribute to data analysis, interpretation, or intellectual content. The authors verified the manuscript for originality and ethical compliance.

Consent to participant

Not applicable.

Consent for Publication

Not applicable. This study used de-identified, publicly available data and does not include any identifiable information.

Conflict of Interest

The authors declare no competing interests.

CRediT authorship contribution statement

Md Salek Miah: Solely responsible for Conceptualization, Data Curation, Methodology, Resources, Project Administration, Formal Analysis, and all stages of Writing – Original Draft, Writing – Review & Editing of the Final Manuscript.

Maria Bintey Kabir: Assisting to Handle of reviewers' comments and Editing of the Draft Manuscript.

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References

Abedin, S., & Arunachalam, D. (2020). Maternal autonomy and high-risk pregnancy in Bangladesh: The mediating influences of childbearing practices and antenatal care. *BMC Pregnancy and Childbirth*, *20*(1). https://doi.org/10.1186/s12884-020-03260-9

Acharya, D. R., Bell, J. S., Simkhada, P., Van Teijlingen, E. R., & Regmi, P. R. (2010a). Open Access RESEARCH Women's autonomy in household decision-making: a demographic study in Nepal. In *Reproductive Health* (Vol. 7). http://www.reproductive-health-journal.com/content/7/1/15

Acharya, D. R., Bell, J. S., Simkhada, P., Van Teijlingen, E. R., & Regmi, P. R. (2010b). Women's autonomy in household decision-making: A demographic study in Nepal. *Reproductive Health*, *7*(1). https://doi.org/10.1186/1742-4755-7-15

Alkema, L., Chou, D., Hogan, D., Zhang, S., Moller, A. B., Gemmill, A., Fat, D. M., Boerma, T., Temmerman, M., Mathers, C., & Say, L. (2016). Global, regional, and national levels and trends in maternal mortality between 1990 and 2015, with scenario-based projections to 2030: A systematic analysis by the un Maternal Mortality Estimation Inter-Agency Group. *The Lancet*, *387*(10017), 462–474. https://doi.org/10.1016/S0140-6736(15)00838-7

Ameyaw, E. K., Tanle, A., Kissah-Korsah, K., & Amo-Adjei, J. (2016). Women's Health Decision-Making Autonomy and Skilled Birth Attendance in Ghana. *International Journal of Reproductive Medicine*, *2016*, 1–9. https://doi.org/10.1155/2016/6569514

Amouzou, A., Barros, A. J. D., Requejo, J., Faye, C., Akseer, N., Bendavid, E., Blumenberg, C., Borghi, J., El Baz, S., Federspiel, F., Ferreira, L. Z., Hazel, E., Heft-Neal, S., Hellwig, F., Liu, L., Maïga, A., Munos, M., Pitt, C., Shawar, Y. R., … Boerma, T. (2025). The 2025 report of the Lancet Countdown to 2030 for women's, children's, and adolescents' health: tracking progress on health and nutrition. *The Lancet*, *405*(10488), 1505–1554. https://doi.org/10.1016/S0140-6736(25)00151-5

Anfaara, F. W., Amoak, D., Kye, N. O., Sano, Y., & Antabe, R. (2024). Exploring the link between household structure and women's household decision-making autonomy in Mauritania. *Journal of Biosocial Science*. https://doi.org/10.1017/S0021932024000221

Antabe, R., Antabe, G., Sano, Y., & Pienaah, C. K. A. (2025). Women's household decision-making autonomy and mental health outcomes in Mozambique. *Global Mental Health*, *12*. https://doi.org/10.1017/gmh.2025.29

Asmare, A. A., Tegegne, A. S., & Belay, D. B. (2026). Unveiling hidden heterogeneity and inequalities in the continuum of care for reproductive, maternal, and child health services in sub-Saharan Africa: A multilevel latent class analysis approach. *Global Epidemiology*, *11*. https://doi.org/10.1016/j.gloepi.2025.100237

Bandara, N. A., Al-Anzi, S. M. F., Zhdanova, A., & Hirani, S. (2024). Women's Empowerment and Mental Health: A Scoping Review. *Women*, *4*(3), 277–289. https://doi.org/10.3390/women4030021

Boakye, S. K., Danquah, L., & Antwi, M. (2025). Maternal mental health improvement in ghana: role of green spaces exposure. *BMC Women's Health*, *25*(1). https://doi.org/10.1186/s12905-025-03886-x

Borges do Nascimento, I. J., Abdulazeem, H. M., Weerasekara, I., Marquez, J., Vasanthan, L. T., Deeken, G., Morgan, R., Tan, H. L., Yordi Aguirre, I., Østeengaard, L., Kularathne, I., Azzopardi-Muscat, N., van Kessel, R., Martinez, E. Z., Permanand, G., & Novillo-Ortiz, D. (2025). Transforming women's health, empowerment, and gender equality with digital health: evidence-based policy and practice. In *The Lancet Digital Health* (Vol. 7, Number 6). Elsevier Ltd. https://doi.org/10.1016/j.landig.2025.01.014

Bushu, G., Kambale, C., Kikwabantu, S., Benedite, F. B., Weber, W., Black, B. O., & Verschueren, K. J. C. (2025). Caesarean section Robson classification, complications and lessons learned in a rural hospital in Walikale, North Kivu, Democratic Republic of Congo: a cross-sectional study. *AJOG Global Reports*, 100586. https://doi.org/10.1016/j.xagr.2025.100586

Celentano ScD MHS, D. D., & Szklo MPH DrPH, M. (2019). *Unlock your eBook today*.

Cuschieri, S. (2019). The STROBE guidelines. In *Saudi Journal of Anaesthesia* (Vol. 13, Number 5, pp. S31–S34). Wolters Kluwer Medknow Publications. https://doi.org/10.4103/sja.SJA_543_18

Dangare, P., Cronje, P. J. R., Mashimbye, Z. E., Masanganise, J., Ntshidi, Z., Gokool, S., Naiken, V., Sawunyama, T., & Dzikiti, S. (2026). Modelling water - use and yield of selected irrigated subtropical crops using machine learning and hybrid models in north - eastern South Africa. *Agricultural Water Management*, *324*. https://doi.org/10.1016/j.agwat.2025.110113

DHS Program, T. (2023). *Lesotho Demographic and Health Survey*.

Esteban-Gonzalo, S., Khan, M. H. A., Hasan, M. M., & Díaz-Morales, J. F. (2026). Psychosocial predictors of women's mental health in Bangladesh: Evidence from BDHS-2022 data on fertility, empowerment, decision-making, child loss, and sex-affective behaviours. *Public Health*, *254*. https://doi.org/10.1016/j.puhe.2026.106222

Fernández, L. G., Yoon, G. H., Firima, E., Gupta, R., Sematle, M. P., Khomolishoele, M., Molulela, M., Bane, M., Tlahani, M., Lee, T. T., Chammartin, F., Gerber, F., Lejone, T., Ayakaka, I., Labhardt, N. D., Amstutz, A., & Belus, J. M. (2025). Prevalence of Mental Health and Substance use Problems and Awareness of Need for Services in Lesotho: Results from a Population-Based Survey. *International Journal of Mental Health and Addiction*, *23*(5), 3607–3626. https://doi.org/10.1007/s11469-024-01309-w

Flor, L. S., Spencer, C. N., Cagney, J., Gil, G. F., Aalruz, H., Abd ElHafeez, S., Abdelwahab, S. I., Abdoun, M., Abebe, M., Abebe, Y., Abedi, A., Abeldaño Zuñiga, R. A., Abie, A., Abiodun, O., Aboagye, R. G., Abreu, L. G., Abu Farha, R. K., Abubakar, B., Abuhammad, S., … Gakidou, E. (2026). Disease burden attributable to intimate partner violence against females and sexual violence against children in 204 countries and territories, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet*, *407*(10523), 31–52. https://doi.org/10.1016/S0140-6736(25)02503-6

Gebeyehu, N. A., Gelaw, K. A., Lake, E. A., Adela, G. A., Tegegne, K. D., & Shewangashaw, N. E. (2022). Women decision-making autonomy on maternal health service and associated factors in low- and middle-income countries: Systematic review and meta-analysis. In *Women's Health* (Vol. 18). SAGE Publications Ltd. https://doi.org/10.1177/17455057221122618

Heise, L., Greene, M. E., Opper, N., Stavropoulou, M., Harper, C., Nascimento, M., Zewdie, D., Darmstadt, G. L., Greene, M. E., Hawkes, S., Henry, S., Heymann, J., Klugman, J., Levine, R., Raj, A., & Rao Gupta, G. (2019). Gender inequality and restrictive gender norms: framing the challenges to health. In *The Lancet* (Vol. 393, Number 10189, pp. 2440–2454). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(19)30652-X

Herrman, H., Patel, V., Kieling, C., Berk, M., Buchweitz, C., Cuijpers, P., Furukawa, T. A., Kessler, R. C., Kohrt, B. A., Maj, M., McGorry, P., Reynolds, C. F., Weissman, M. M., Chibanda, D., Dowrick, C., Howard, L. M., Hoven, C. W., Knapp, M., Mayberg, H. S., … Wolpert, M. (2022). Time for united action on depression: a Lancet–World Psychiatric Association Commission. *The Lancet*, *399*(10328), 957–1022. https://doi.org/10.1016/S0140-6736(21)02141-3

Hindman, J., Zugai, J., Foster, K., & Raeburn, T. (2026). Self-Determination Theory as a Framework for Research and Design of Digital Applications for Nurses' Well-Being. In *Journal of Advanced*

Nursing (Vol. 82, Number 1, pp. 897–906). John Wiley and Sons Inc. https://doi.org/10.1111/jan.16895

Hormazábal-Salgado, R., Whitehead, D., Osman, A. D., & Hills, D. (2024). Person-Centred Decision-Making in Mental Health: A Scoping Review. In *Issues in Mental Health Nursing* (Vol. 45, Number 3, pp. 294–310). Taylor and Francis Ltd. https://doi.org/10.1080/01612840.2023.2288181

Idris, I. B., Hamis, A. A., Bukhori, A. B. M., Hoong, D. C. C., Yusop, H., Shaharuddin, M. A. A., Fauzi, N. A. F. A., & Kandayah, T. (2023). Women's autonomy in healthcare decision making: a systematic review. *BMC Women's Health*, *23*(1). https://doi.org/10.1186/s12905-023-02792-4

Jejaw, M., Kelkay, J. M., Demissie, K. A., Hagos, A., Teshale, G., Baykemagn, N. D., Tadele, A., Tafere, T. Z., Aschalew, A. Y., & Yehuala, T. Z. (2026). Estimated Impact of Pressure to Conceive on Maternal Depression in Lesotho: A Quasi-experimental Propensity Score Matching using DHS Data. *Archives of Women's Mental Health*, *29*(2). https://doi.org/10.1007/s00737-026-01687-1

Jia, J., Snider, A., Collishaw, A., McNamara, P. E., & Tumusiime, E. (2025). Pathways linking WASH access and women's empowerment: Evidence from Zambia and Honduras. *Journal of Rural Studies*, *116*. https://doi.org/10.1016/j.jrurstud.2025.103602

Kamiya, Y. (2011). Women's autonomy and reproductive health care utilisation: Empirical evidence from Tajikistan. *Health Policy*, *102*(2–3), 304–313. https://doi.org/10.1016/j.healthpol.2011.04.001

Leight, J., Pedehombga, A., Ganaba, R., & Gelli, A. (2022a). Women's empowerment, maternal depression, and stress: Evidence from rural Burkina Faso. *SSM - Mental Health*, *2*. https://doi.org/10.1016/j.ssmmh.2022.100160

Leight, J., Pedehombga, A., Ganaba, R., & Gelli, A. (2022b). Women's empowerment, maternal depression, and stress: Evidence from rural Burkina Faso. *SSM - Mental Health*, *2*. https://doi.org/10.1016/j.ssmmh.2022.100160

Leseba, N., Vermaak, K., Makatjane, T., & Lebuso, M. (2025). A multilevel analysis of factors associated with stunting among children under five years in Lesotho: a study of the lesotho multiple cluster indicator survey 2018. *Journal of Health, Population and Nutrition*, *44*(1). https://doi.org/10.1186/s41043-025-00901-7

Llop-Gironés, A., Julià, M., Chicumbe, S., Dulá, J., Odallah, A. A. P., Alvarez, F., Zahinos, I., Mazive, E., & Benach, J. (2019). Inequalities in the access to and quality of healthcare in Mozambique: Evidence from the household budget survey. *International Journal for Quality in Health Care*, *31*(8), 577–582. https://doi.org/10.1093/intqhc/mzy218

Melkam, M., Fente, B. M., Negussie, Y. M., Asmare, Z. A., Asebe, H. A., Seifu, B. L., Gebrehana, A. K., Tsega, S. S., Bezie, M. M., & Asnake, A. A. (2025). Postpartum depression and associated factors among childbearing women from the recent Demographic and Health Survey data of Mozambique: Multilevel analysis. *Heliyon*, *11*(1). https://doi.org/10.1016/j.heliyon.2024.e41106

Miah, M. S., Lamia, F. F., Begum, Mst. L., Shied, M., & Akhter, R. (2026). Household Water and Handwashing Facilities and Early Childhood Education Participation in Nepal: Evidence From a

Nationally Representative Cross-Sectional Survey. *Health Science Reports*, *9*(4). https://doi.org/10.1002/hsr2.72254

Miah, M. S. U. M. O. (2025). *Associations of women's empowerment and intimate partner violence with mental health among ever-married women in Bangladesh*.

Miah, M. S., & Ullah, M. O. (2025). Associations of internet use and pregnancy loss with depression and anxiety among women in Bangladesh: evidence from the 2022 BDHS. *BMC Women's Health*. https://doi.org/10.1186/s12905-025-04166-4

Miller, S., Abalos, E., Chamillard, M., Ciapponi, A., Colaci, D., Comandé, D., Diaz, V., Geller, S., Hanson, C., Langer, A., Manuelli, V., Millar, K., Morhason-Bello, I., Castro, C. P., Pileggi, V. N., Robinson, N., Skaer, M., Souza, J. P., Vogel, J. P., & Althabe, F. (2016). Beyond too little, too late and too much, too soon: a pathway towards evidence-based, respectful maternity care worldwide. In *The Lancet* (Vol. 388, Number 10056, pp. 2176–2192). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(16)31472-6

Moeti, M., Gao, G. F., & Herrman, H. (2022). Global pandemic perspectives: public health, mental health, and lessons for the future. In *The Lancet* (Vol. 400, Number 10353, pp. e3–e7). Elsevier B.V. https://doi.org/10.1016/S0140-6736(22)01328-9

Mokwena, K. E., & Makhozonke, N. (2024). The prevalence of postnatal depression and associated factors among clinic attendees in Maseru, Lesotho. *International Journal of Innovative Research and Scientific Studies*, *7*(2), 398–408. https://doi.org/10.53894/ijirss.v7i2.2644

Mustafiz, R. B., Chowdhury, S. S. A., Kundu, S., Munaf, N. B., & Sharif, A. Bin. (2025). Determinants of healthcare decision-making autonomy among Bangladeshi women: mixed-effect logistic regression analysis. *BMC Women's Health*, *25*(1). https://doi.org/10.1186/s12905-025-03666-7

Myint, W. W., Yusuf, A., Samman, E., Nguyen, A., Mendez, S. E., & Clark, H. R. (2025a). Factors associated with severity of depression among women intimate partner violence survivors in Lesotho: A cross-sectional study. *Women's Health*, *21*. https://doi.org/10.1177/17455057251387843

Myint, W. W., Yusuf, A., Samman, E., Nguyen, A., Mendez, S. E., & Clark, H. R. (2025b). Factors associated with severity of depression among women intimate partner violence survivors in Lesotho: A cross-sectional study. *Women's Health*, *21*. https://doi.org/10.1177/17455057251387843

Nabaggala, M. S., Reddy, T., & Manda, S. (2021). Effects of rural–urban residence and education on intimate partner violence among women in Sub-Saharan Africa: a meta-analysis of health survey data. *BMC Women's Health*, *21*(1). https://doi.org/10.1186/s12905-021-01286-5

Noor, S. T. A., Siddique, S., Das, O., Yeasar, S., & Islam, R. B. (2025). Exploring mental health disparities in Mozambique: Depression and anxiety symptoms among reproductive-aged women using data from Mozambique Demographic and Health Survey 2022–23. *Global Epidemiology*, *10*. https://doi.org/10.1016/j.gloepi.2025.100223

Noor, S. T. A., Siddique, S., Yeasar, S., Das, O., Tanvir, S.-U.-A., & Islam, R. B. (2025). *Prevalence, Determinants, and Care-Seeking Behaviour for Major Depressive Disorder in the Lesotho*

Population: A Multilevel Analysis from the Lesotho Demographic and Health Survey 2023-24. https://doi.org/10.21203/rs.3.rs-6945317/v1

Okyere, J., Ayebeng, C., & Dickson, K. S. (2025a). Sex differences in help-seeking behavior for depression in Lesotho: findings from a national survey. *BMC Psychiatry*, *25*(1). https://doi.org/10.1186/s12888-025-06749-9

Okyere, J., Ayebeng, C., & Dickson, K. S. (2025b). Sex differences in help-seeking behavior for depression in Lesotho: findings from a national survey. *BMC Psychiatry*, *25*(1). https://doi.org/10.1186/s12888-025-06749-9

Oram, S., Fisher, H. L., Minnis, H., Seedat, S., Walby, S., Hegarty, K., Rouf, K., Angénieux, C., Callard, F., Chandra, P. S., Fazel, S., Garcia-Moreno, C., Henderson, M., Howarth, E., Macmillan, H. L., Murray, L. K., Othman, S., Robotham, D., Rondon, M. B., … Howard, L. M. (2022). *The Lancet Psychiatry Commissions The Lancet Psychiatry Commission on intimate partner violence and mental health: advancing mental health services, research, and policy* (Vol. 9). www.thelancet.com/psychiatry

Ostovar, A., Davari, D. D., & Dzikuć, M. (2025). Determinants of Design with Multilayer Perceptron Neural Networks: A Comparison with Logistic Regression. *Sustainability (Switzerland)*, *17*(6). https://doi.org/10.3390/su17062611

Pandey, A. R., Adhikari, B., Bista, B., Lamichhane, B., Joshi, D., Saugat Pratap, K. C., Sharma, S., & Baral, S. (2024). Prevalence, determinants and care-seeking behaviour for anxiety and depression in Nepalese population: a secondary analysis of data from Nepal Demographic and Health Survey 2022. *BMJ Open*, *14*(8). https://doi.org/10.1136/bmjopen-2023-078582

Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., Chisholm, D., Collins, P. Y., Cooper, J. L., Eaton, J., Herrman, H., Herzallah, M. M., Huang, Y., Jordans, M. J. D., Kleinman, A., Medina-Mora, M. E., Morgan, E., Niaz, U., Omigbodun, O., … UnÜtzer, Jü. (2018a). The Lancet Commission on global mental health and sustainable development. In *The Lancet* (Vol. 392, Number 10157, pp. 1553–1598). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(18)31612-X

Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., Chisholm, D., Collins, P. Y., Cooper, J. L., Eaton, J., Herrman, H., Herzallah, M. M., Huang, Y., Jordans, M. J. D., Kleinman, A., Medina-Mora, M. E., Morgan, E., Niaz, U., Omigbodun, O., … UnÜtzer, Jü. (2018b). The Lancet Commission on global mental health and sustainable development. In *The Lancet* (Vol. 392, Number 10157, pp. 1553–1598). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(18)31612-X

Quenby, S., Gallos, I. D., Dhillon-Smith, R. K., Podesek, M., Stephenson, M. D., Fisher, J., Brosens, J. J., Brewin, J., Ramhorst, R., Lucas, E. S., McCoy, R. C., Anderson, R., Daher, S., Regan, L., Al-Memar, M., Bourne, T., MacIntyre, D. A., Rai, R., Christiansen, O. B., … Coomarasamy, A. (2021). Miscarriage matters: the epidemiological, physical, psychological, and economic costs of early pregnancy loss. In *The Lancet* (Vol. 397, Number 10285, pp. 1658–1667). Elsevier B.V. https://doi.org/10.1016/S0140-6736(21)00682-6

Rahman, T., Gasbarro, D., Alam, K., & Alam, K. (2024). Rural–urban disparities in household catastrophic health expenditure in Bangladesh: a multivariate decomposition analysis. *International Journal for Equity in Health*, *23*(1). https://doi.org/10.1186/s12939-024-02125-3

Raz, S., Rahman, M. S., Aktar, B., & Rashid, S. F. (2025). Women's household decision-making autonomy in low economic settlements in Bangladesh: Evidence from a cross-sectional study, 2022? *Women's Studies International Forum*, *113*. https://doi.org/10.1016/j.wsif.2025.103181

Russell, L. B., Sobanjo-ter Meulen, A., & Toscano, C. M. (2021). Evaluating the cost-effectiveness of maternal pertussis immunization in low- and middle-income countries: A review of lessons learnt. In *Vaccine* (Vol. 39, Number 1, pp. 121–124). Elsevier Ltd. https://doi.org/10.1016/j.vaccine.2020.10.054

Tamanna, T., & Mahmud, S. (2025). Applying machine learning to predict quality ANC determinants in Bangladesh: a BDHS-2022 cross-sectional study. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-21411-w

Tamanna, T., Mahmud, S., Salma, N., Hossain, M. M., & Karim, M. R. (2025). Identifying determinants of malnutrition in under-five children in Bangladesh: insights from the BDHS-2022 cross-sectional study. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-99288-y

The Lancet. (2025). Empowering women during childbirth. In *The Lancet* (Vol. 406, Number 10498, p. 1). Elsevier B.V. https://doi.org/10.1016/S0140-6736(25)01382-0

Thibaut, F., & ELNahas, G. (2023). Women's Mental Health and Lessons Learnt from the COVID-19 Pandemic. In *Psychiatric Clinics of North America* (Vol. 46, Number 3, pp. 415–426). W.B. Saunders. https://doi.org/10.1016/j.psc.2023.04.001

Viens, L. J., Clouston, S., & Messina, C. R. (2016). Women's autonomy and cervical cancer screening in the Lesotho Demographic and Health Survey 2009. *Social Science and Medicine*, *150*, 23–30. https://doi.org/10.1016/j.socscimed.2015.12.009

Westby, C. L., Erlandsen, A. R., Nilsen, S. A., Visted, E., & Thimm, J. C. (2021). Depression, anxiety, PTSD, and OCD after stillbirth: a systematic review. *BMC Pregnancy and Childbirth*, *21*(1). https://doi.org/10.1186/s12884-021-04254-x

Zhang, Y., Augusto, O., Ásbjörnsdóttir, K., Akullian, A., Cumbe, V., Rao, D., Gimbel, S., Sherr, K., Wagenaar, B. H., Manuel, J. L., Hicks, L., Mahumane, A., Pfeiffer, J., Gloyd, S., Cuembelo, F., Nhumba, M., Lequechane, J., Covele, A., Napua, M., … Michel, C. (2019). Geographic distribution and determinants of mental health stigma in central Mozambique. *Social Psychiatry and Psychiatric Epidemiology*, *54*(11), 1391–1410. https://doi.org/10.1007/s00127-019-01708-8

Zoha, S., Alam, S., Sifat, I. K., Sultana, N., & Kibria, M. K. (2025). Identifying determinants and predicting cesarean section delivery among Bangladeshi women using machine learning: Insight from BDHS 2022 Data. *PLOS Global Public Health*, *5*(11 November). https://doi.org/10.1371/journal.pgph.0005494

List of All Figures

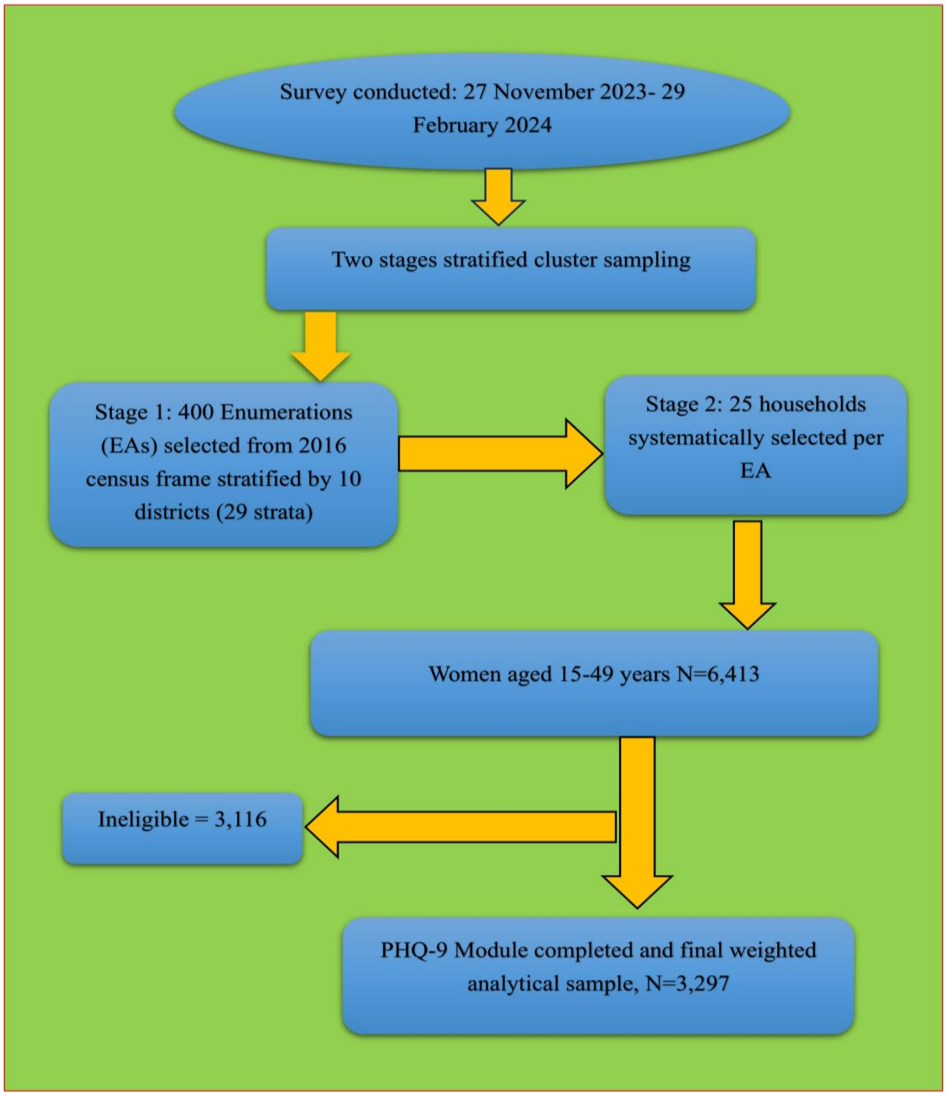

Figure 7: The Study Flow

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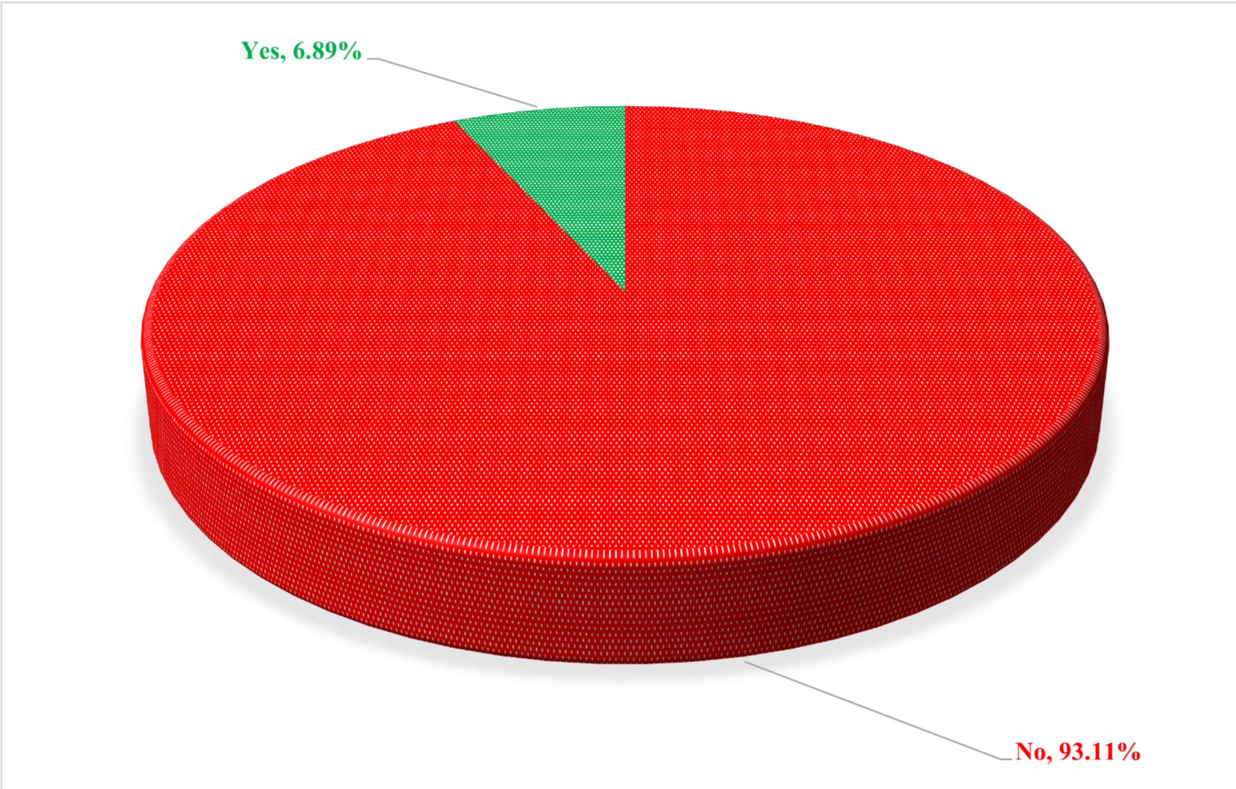

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Figure 8 : Prevalences of Depression Symptoms

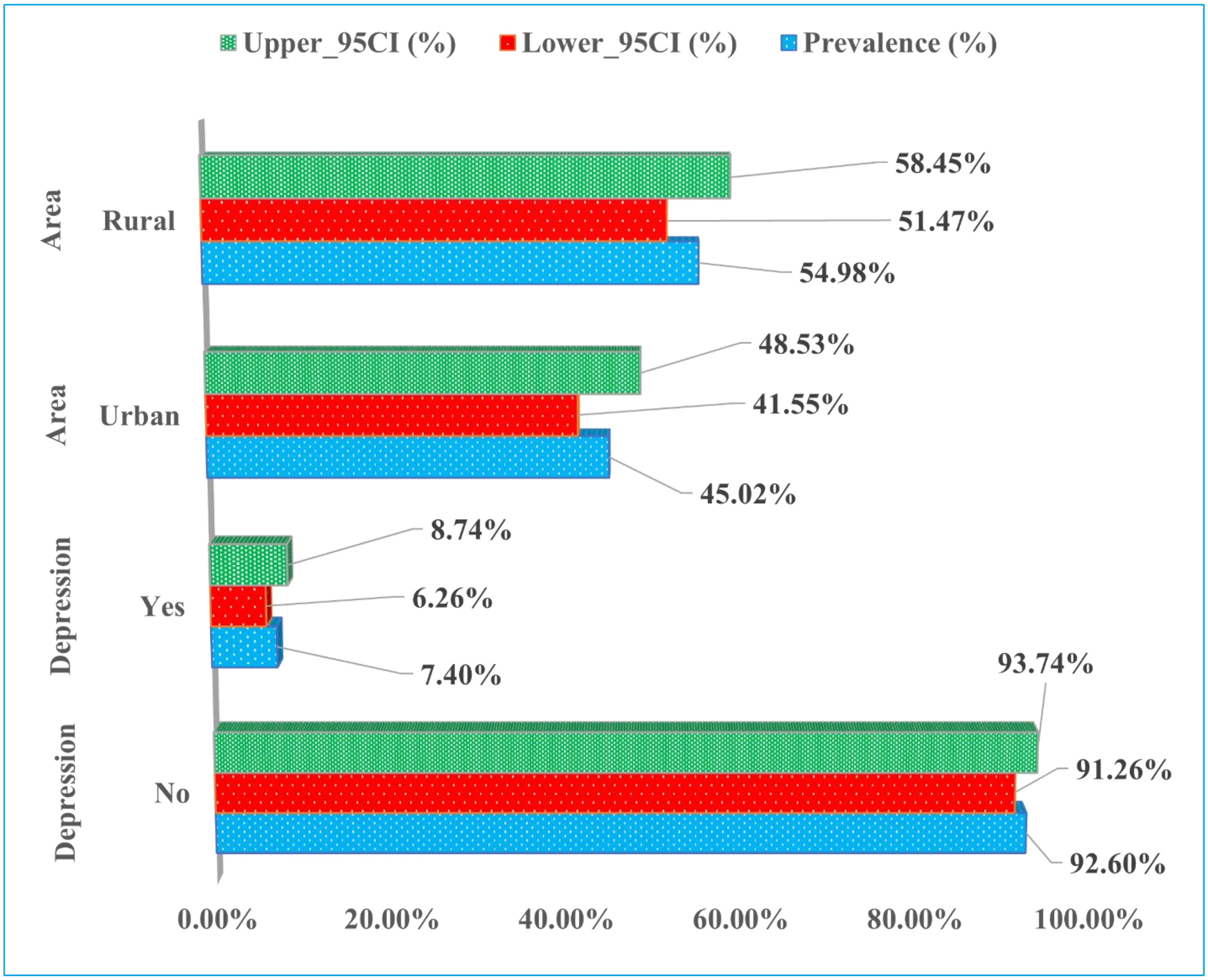

Figure 9: Rural-Urban Inequalities of Depression Prevalence

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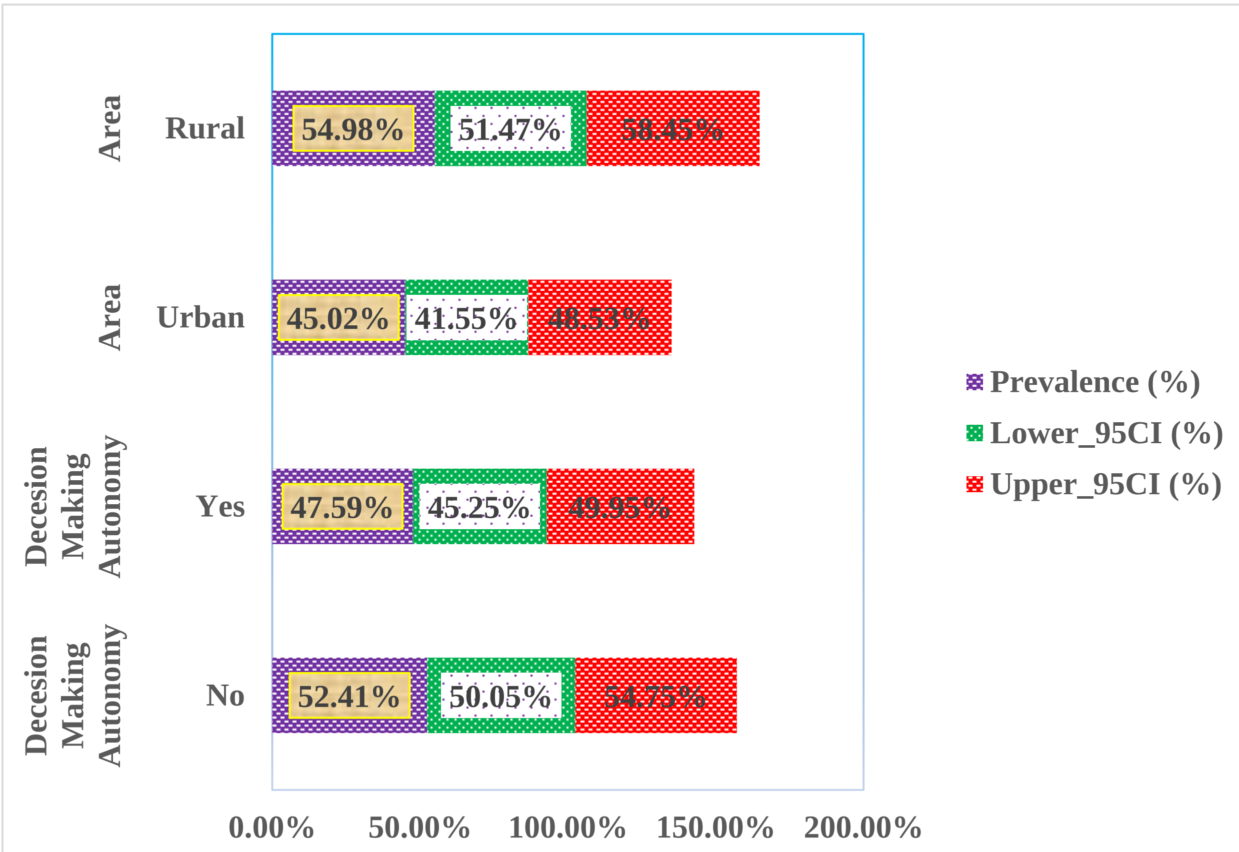

Figure 10: Rural-Urban Inequalities of Decision -Making Autonomy

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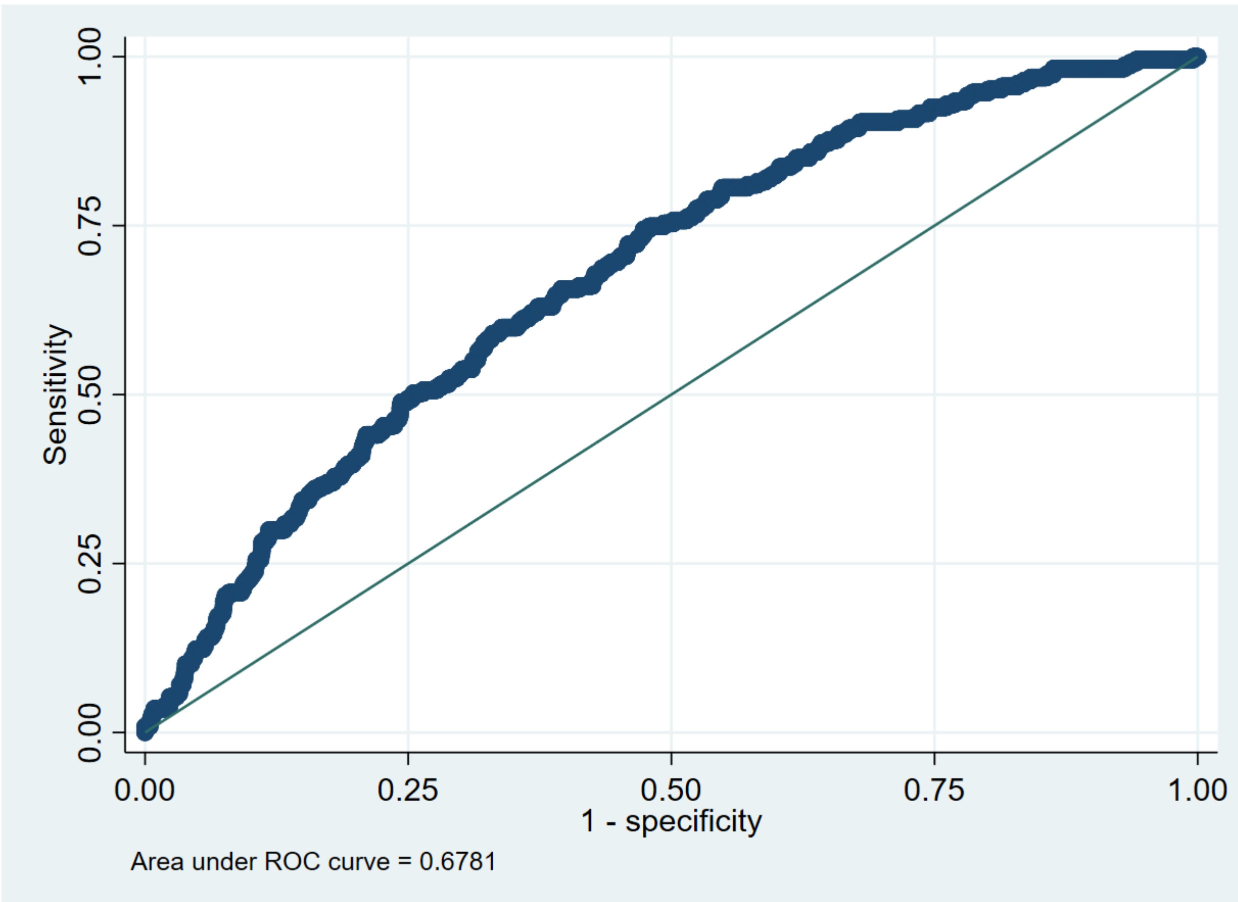

Figure 11:ROC curve showing model performance

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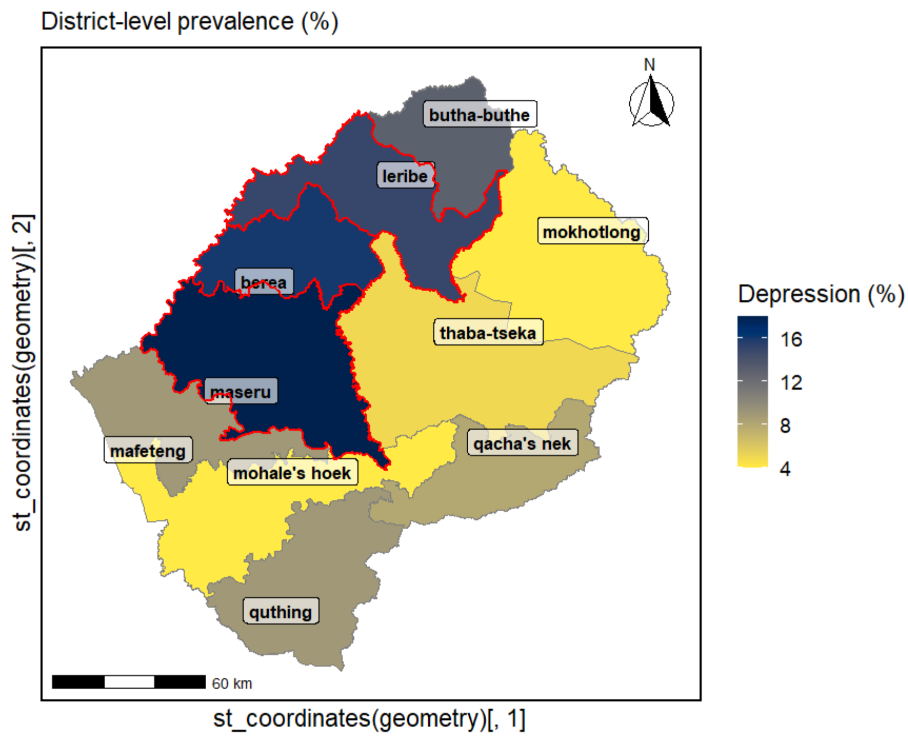

Figure 12: Spatial Inequalities of Depression Prevalences

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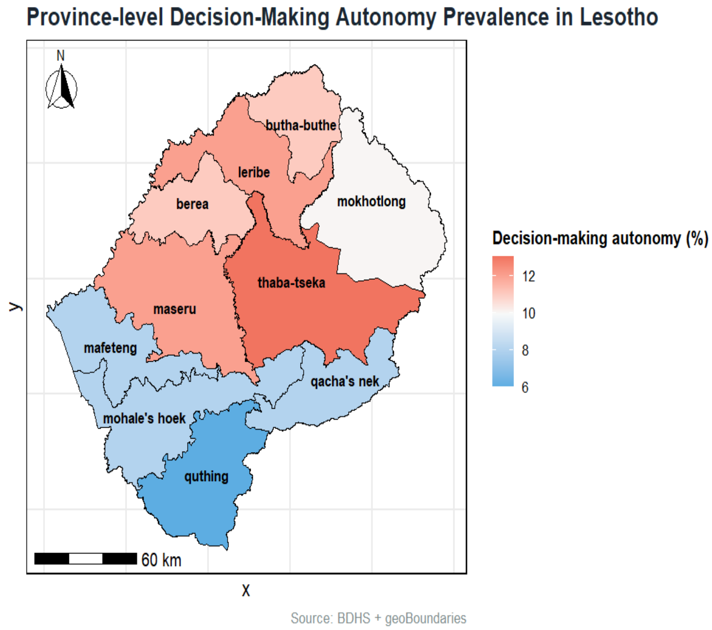

Figure 13: Spatial Inequalities of Decision-making Autonomy

Table 3: Distribution of depressive symptoms (PHQ-9 ≥10) by sociodemographic and reproductive characteristics (N=3,297)

Characteristics	Depressive Symptoms (>=10) %, (95% CI)	
	No	Yes
Overall	92.6 (91.3, 93.7)	7.4 (6.3, 8.7)
Women's age		
15-19	19.9 (18.2, 21.7)	19.6 (13.4, 27.8)
20-24	18.4 (16.7, 20.3)	14.5 (9.3, 21.8)
25-29	13.5 (11.9, 15.3)	16.2 (10.1, 24.9)
30-34	13.6 (12.2, 15.1)	13.2 (8.4, 20.1)
35-39	13.1 (11.6, 14.9)	16.6 (11.1, 24.1)
40-44	11.9 (10.4, 13.6)	13.5(8.9, 20.0)
45-49	9.6 (8.0, 11.5)	6.5 (3.7, 11.1)
Age at first birth		
10-17 (Early)	11.4 (9.6, 13.4)	17.1 (11.6, 24.5)
18-24 (Normal)	44.0 (41.8, 46.3)	45.8 (38.1, 53.7)
≥25 (Late)	44.6 (42.4, 46.8)	37.0 (29.9, 44.8)
Age at first sexual intercourse		
<18	54.4 (51.6, 57.2)	64.0 (56.4, 70.9)
≥18	45.6 (42.8, 48.4)	36.0 (29.1, 43.6)
Age at cohabitation		
<18 (Young)	14.2 (12.4, 16.1)	19.4 (13.9, 26.3)
18–20 (Mild)	20.2 (18.5, 22.0)	17.0 (11.7, 24.0)
≥21 (Older)	65.6 (63.3, 67.9)	63.7 (55.2, 71.3)
Women's education		
Below Primary	24.7 (22.1, 27.4)	28.7 (21.0, 38.0)
Secondary incomplete	59.2 (56.5, 61.9)	54.1 (45.3, 62.6)
Secondary or higher	16.1 (13.3, 19.5)	17.2 (11.4, 25.2)
Maternal Employment		
Not working	61.0 (57.7, 64.2)	61.6 (52.9, 69.6)
Currently working	39.0 (35.8, 42.3)	38.4 (30.4, 47.1)
Adequate ANC Visits		
<4 visits	81.0 (79.1, 82.8)	82.5 (75.5, 87.9)
≥4 visits	19.0 (17.2, 20.9)	17.5 (12.1, 24.5)
Number of children		
None	33.6 (31.5, 35.8)	26.7 (20.4, 34.2)
1-2	48.2 (45.5, 51.0)	48.5 (40.9, 56.1)
3+	18.2 (16.4, 20.0)	24.8 (18.8, 32.1)
Birth Size		
Small	81.5 (79.6, 83.2)	78.7 (71.3, 84.7)
Average or Large	18.5 (16.8, 20.4)	21.3 (15.3, 28.7)
Preterm birth		
Term (>=9 months)	91.8 (90.1, 93.2)	90.9 (84.8, 94.7)

Preterm (<9 months)	8.2 (6.8, 9.9)	9.1 (5.3, 15.2)
Currently Pregnant		
No	97.1 (96.2, 97.7)	98.0 (94.7, 99.2)
Yes	2.9 (2.3, 3.8)	2.0 (0.8, 5.3)
Pregnancy Status		
Single pregnancy	67.7 (65.5, 69.8)	75.3 (68.8, 81.0)
Multiple pregnancy	32.3 (30.2, 34.5)	24.7 (19.2, 31.2)
Pressure to be pregnant		
No	93.3 (91.5, 94.7)	86.3 (76.2, 92.6)
Yes	6.7 (5.3, 8.5)	13.7 (7.4, 23.8)
History of pregnancy loss		
None	89.7 (88.0, 91.1)	84.5 (78.1, 89.3)
One loss	8.2 (6.8, 9.8)	12.6 (8.2, 18.8)
Two or more losses	2.2 (1.6, 3.0)	2.9 (1.3, 6.7)
Residing Status		
Not residing with husband/partner	35.5 (32.0, 39.2)	37.2 (26.6, 49.2)
Residing with husband/partner	64.5 (60.8, 68.0)	62.8 (50.8, 73.4)
Currently Abstaining		
No	93.6 (92.5, 94.5)	94.6 (90.4, 97.1)
Yes	6.4 (5.5, 7.5)	5.4 (2.9, 9.6)
Number of Unions		
Once	60.5 (58.0, 62.9)	57.3 (48.0, 66.2)
More than once	39.5 (37.1, 42.0)	42.7 (33.8, 52.0)
Menstruated last 6 weeks		
No	24.7 (22.8, 26.8)	27.5 (20.6, 35.8)
Yes	75.3 (73.2, 77.2)	72.5 (64.2, 79.4)
Decision Making Autonomy		
No	52.1 (49.7, 54.5)	56.6 (47.0, 65.7)
Yes	47.9 (45.5, 50.3)	43.4 (34.3, 53.0)
IPV Justification		
Does not justify IPV	80.2 (77.8, 82.5)	83.9 (77.7, 88.6)
Justifies IPV	19.8 (17.5, 22.2)	16.1 (11.4, 22.3)
Household Sanitation facility		
Unimproved	35.4 (32.2, 38.8)	30.6 (23.1, 39.3)
Improved	64.6 (61.2, 67.8)	69.4 (60.7, 76.9)
Source of drinking water		
Unimproved source	16.2 (13.7, 19.0)	13.1 (8.8, 19.1)
Improved source	83.8 (81.0, 86.3)	86.9 (80.9, 91.2)
Place of residence		
Urban	44.3 (40.7, 47.9)	54.5 (46.1, 62.7)
Rural	55.7 (52.1, 59.3)	45.5 (37.3, 53.9)
Administrative province		
Butha-Buthe	6.2 (5.3, 7.2)	6.7 (4.5, 9.8)

Leribe	17.8 (15.4, 20.5)	19.5 (14.1, 26.4)
Berea	15.0 (12.0, 18.6)	18.4 (12.3, 26.6)
Maseru	32.8 (28.8, 37.0)	36.8 (28.3, 46.2)
Mafeteng	6.5 (5.5, 7.5)	5.3 (3.5, 8.0)
Mohale's Hoek	4.7 (4.0, 5.6)	2.2 (1.1, 4.5)
Quthing	3.6 (3.1, 4.3)	3.7 (2.3, 5.8)
Qacha's Nek	2.9 (2.4, 3.5)	2.9 (1.6, 5.2)
Mokhotlong	4.4 (3.5, 5.4)	1.4 (0.7, 2.7)
Thaba-Tseka	6.2 (5.4, 7.1)	3.1 (1.7, 5.6)
Religion		
Other	65.3 (62.9, 67.6)	63.3 (54.5, 71.4)
Roman Catholic	34.7 (32.4, 37.1)	36.7 (28.6, 45.5)
Currently working		
No	61.0 (57.7, 64.2)	61.6 (52.9, 69.6)
Yes	39.0 (35.8, 42.3)	38.4 (30.4, 47.1)
Wealth Status		
Poor	31.3 (27.6, 35.2)	24.9 (19.0, 31.9)
Middle class	19.2 (16.9, 21.7)	19.1 (13.2, 26.9)
Rich	49.4 (45.2, 53.6)	55.9 (47.1, 64.3)
Household Size		
1-3 members	28.3 (25.6, 31.1)	21.7 (15.9, 28.9)
4 or more members	71.6 (68.8, 74.3)	78.2 (71.0, 84.0)
Household head sex		
Male	57.6 (54.9, 60.2)	55.4 (45.6, 64.8)
Female	42.3 (39.7, 45.0)	44.5 (35.1, 54.3)
Household assets		
No	51.3 (47.3, 55.3)	49.2 (41.6, 56.8)
Yes	48.6 (44.6, 52.6)	50.7 (43.1, 58.3)
Media Access		
No	5.7 (4.77, 6.90)	3.2 (1.79, 5.83)
Yes	94.2 (93.0, 95.2)	96.7 (94.1, 98.2)
Internet use status		
No internet use	19.6 (17.7, 21.6)	22.0 (14.7, 31.4)
Internet use last 12 months	80.3 (78.3, 82.2)	77.9 (68.5, 85.2)
Household materials		
Unimproved	17.0 (14.9, 19.2)	11.5 (7.8, 16.7)
Improved	82.9 (80.7, 85.0)	88.4 (83.2, 92.1)

Table 4:Associations Between Decision Making Autonomy and Depression Symptoms (N= 3,297, N1 Rural= 2,086, N2 Urban =1,211)

Characteristics	Depression (PHQ-9 ≥10)			
	COR (95% CI), p	AOR (95% CI),p		
Decision Making Autonomy	-	**National Level**	**Rural**	**Urban**
No (Ref)	1	1	1	1
Yes	0.83 0.56-1.23) ,0.368	0.63(0.39-0.98), 0.048	0.44(0.27-0.74) ,0.002	0.88 (0.39-1.97) ,0.758

Note: OR = odds ratio; CI = confidence interval; PHQ-9 ≥10 indicates depression; GAD-7 ≥10 indicates anxiety.

Confounders: women age, women education, maternal employment, religion, administrative division, age at first birth, age at cohabitation, number of children, pregnancy status, pregnancy loss, place of residence, toilet facilities, water access, household head sex, household size category, household wealth status, media access, internet use, household assets, and household materials

Decision-Making Autonomy and Depressive Symptoms Among Women in Lesotho: Rural–Urban Differences

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Abstract

Objectives:

To assess the association between depressive symptoms and decision-making autonomy with rural urban inequalities, and mapped province-level spatial disparities in depression prevalence in ever-married women in Lesotho.

Methods:

This study used cross-section data of 3,297 women aged 15–49 years from the 2023–24 Lesotho Demographic and Health Survey (LDHS). Depression was measured using the Patient Health Questionnaire-9 (PHQ-9 ≥10). Women's decision-making autonomy was measured using a composite empowerment index based on participation in five household decision-making domains. Survey-weighted multivariable logistic regression models were fitted at the national level and stratified by place of residence. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC). Spatial analyses were conducted to examine district-level variation in depression prevalence and autonomy.

Results:

Overall, 7.4% (95% CI: 6.3–8.7) of women reported ~~moderate-to-severe~~moderate-severe depressive symptoms. Women with decision-making autonomy showed lower odds of depression nationally (AOR = 0.63; 95% CI: 0.39–0.98), with a stronger effect in rural areas (AOR = 0.44; 95% CI: 0.27–0.74). Depression prevalence was highest in central–northern districts (Maseru, Berea, Leribe, Butha-Buthe) and lower levels of autonomy observed ~~highlighted~~ in southern and south-western provinces.

Conclusions:

Women's decision-making autonomy is significantly associated with lower odds of depressive symptoms in Lesotho, particularly among rural women. The observed~~detected~~ rural–urban and spatial inequalities highlight the need for context-specific mental health and reproductive health interventions that incorporate women's empowerment as a core component.

Keywords: Depression, Decision-making autonomy, Pregnancy loss, Rural–urban disparities, Lesotho, Spatial analysis

Highlights

- Depression affected 7.4% of women, with marked rural–urban and geographic inequalities.
- Higher decision-making autonomy was associated with lower odds of depression, particularly in rural areas.
- Spatial clustering revealed high-burden districts, indicating the need for targeted, context-specific interventions.

Introduction

Depression is one of the causes of disability in the whole world and it is even higher among women in reproductive age (Herrman et al., 2022). Depression is more common and prevalent in women than men because of an aggregate of biological, social, and environmental issues (Patel et al., 2018a). Social roles are gendered, and access to economic resources is limited in low-resource settings, making people more vulnerable to mental health problems due to reproductive roles and restricted access to mental health (Patel et al., 2018b).

Women possessing depression do not only impact the quality of life, but the functioning of the family, child health, and well-being of the community, in general (Oram et al., 2022). It is thus important to understand the social determinants that affect the depressive symptoms in order to bring meaningful, context-sensitive intervention (Amouzou et al., 2025). The decision-making autonomy of women is one of the important social determinants of mental health (The Lancet, 2025).

Decision-making autonomy is defined~~scribed~~ as a woman having power to engage in making decisions regarding her health care, family purchases, finances, mobility and reproductive decisions (Heise et al., 2019). The results of numerous nations show that women who have more freedom are less likely to encounter depressive and anxiety symptoms (Westby et al., 2021). Autonomy can keep mental health safe because it would enhance the sense of control that women have in their lives, decrease their vulnerability to coercive relationships, and allow access to health and social resources (Leight et al., 2022a). On the other hand, a lack of autonomy has been associated with higher psychological distress, helplessness and increased vulnerability to common mental disorders (Leight et al., 2022b; M. S. U. M. O. Miah, 2025). Although the increased awareness of the need of autonomy among women and the mental wellness of women has advanced throughout the globe, few studies have been conducted in most low-income countries (LMICs) (Borges do Nascimento et al., 2025; Jia et al., 2025). Mental health of women in sub-Saharan Africa is a product of socioeconomic, reproductive, and contextual risk factors (Bandara et al., 2024). There are early marriage, early childbearing, internet use, pregnancy loss, high fertility and large family size, which are usually stressful factors that can make an individual vulnerable to depression (M. S. Miah & Ullah, 2025). These risk factors are usually accompanied by limited decision-making autonomy especially in rural regions where traditions and gender norms are the order of the day (Rahman et al., 2024). Although urban setting usually has more access to education, work, and health services, other stressors, such as economic insecurity, social isolation, and accelerated sociocultural change, might also be brought about by urban setting (Nabaggala et al., 2021). These aspects indicate that the interdependence between autonomy and depression might depend on the setting and there is a necessity to explore the rural-urban variations in mental health outcomes (Acharya et al., 2010b, 2010a; Idris et al., 2023; Mustafiz et al., 2025).

In Southern Africa, in Lesotho, depressive symptoms remain a major public health issue, with recent national surveys reporting a substantial burden of untreated mental health conditions among women, driven by digital access, poverty, human immunodeficiency virus (HIV) prevalence, and limited access to mental-health services (DHS Program, 2023; Jejaw et al., 2026). Additionally, Lesotho showed the highest estimated prevalence of depression, with women excessively suffered (Myint et al., 2025a). Likely, a slightly higher prevalence of women sought (17.1%) help for depression compared to men (16.4%). Particularly, women's living in urban's settings showed higher help-seeking prevalences compared to men (Fernández et al., 2025; Okyere et al., 2025a). Previous studies also shows that women's autonomy in Lesotho remains remarkably inadequate, mainly in decisions related to health care, household purchases, and mobility, with rural women experiencing even lower levels of intervention due to entrenched gender norms and economic dependence (Gebeyehu et al., 2022; Hormazábal-Salgado et al., 2024; Viens et al., 2016).

In Lesotho, limited studies exist about the mental health of women and their social determinants (DHS Program, 2023). There has been a preponderance of studies which have concentrated on maternal and reproductive health outcomes and little has been done on psychological well-being

(Leseba et al., 2025). Naturally, there is scanty evidence on the association~~relationship~~ between decision-making autonomy with depressive symptoms among women in national representative samples (Mokwena & Makhozonke, 2024). In addition, the rural-urban ~~di~~disparities~~fferences~~ or province-wise spatial inequalities ~~variation~~ in the prevalence of depression have not been investigated in previous studies, which are important in determining the areas of high burden and effective intervention strategies (Okyere et al., 2025b). This gap in evidence is a major shortcoming since there is a need to combine policy and programmatic planning in Lesotho to address the issues of women empowerment and mental health (Myint et al., 2025b).

Theoretically, women's autonomy influences mental-health outcomes through multiple mechanisms. Guided by Self-Determination Theory and Gender and Power Theory, limited autonomy reduces women's capacity to make independent decisions regarding health care, household resources, and mobility. Such constraints increase psychological stress, diminish perceived control, weaken coping ability, and ultimately elevate the risk of depressive symptoms (Hindman et al., 2026; Raz et al., 2025). These theoretical perspectives support the link between autonomy and depression in the context of Lesotho (DHS Program, 2023).

Against this background, the present study aimed to examine the association between decision making autonomy and ~~moderate to severe~~moderate-severe depressive symptoms among women in Lesotho using nationally representative LDHS data (DHS Program, 2023). Specifically, this study assessed this association at the national level, conducted stratified analyses by rural and urban residence to explore inequalities, and mapped province-wise spatial variation in depression prevalence. By integrating individual-level regression analyses with spatial methods, this study seeks to provide policy-relevant evidence to inform targeted mental health and reproductive health interventions in Lesotho.

Methodology

Setting, study design and source of data

This paper is based on a secondary analysis of the data of the 2023-24 Lesotho Demographic and Health Survey (LDHS), a household survey that was conducted nationally by the Lesotho Ministry of Health with technical assistance of ICF under the DHS Program. The data was collected from 27 November 2023 to 29 February 2024. To present current estimates of the demographic and health variables, such as fertility, maternal and child health, and mental health according to the Sustainable Development Goals (SDGs). The LDHS utilized standardized DHS questionnaires (household, women, men and biomarker) that were translated to the Lesotho setting and were conducted in the Sesotho and English languages. Face to face interviews were conducted using tablet-based computer aided personal interviewing (CAPI) and inbuilt quality checks were used (DHS Program, 2023).This study followed Strengthening the Reporting of Observational Studies in Epidemiology -(STROBE) guidelines (Cuschieri, 2019).

Sampling design and population of study

LDHS used a stratified cluster sampling design, which was two-stage using the 2016 Lesotho Population and Housing Census. At the first level, 400 enumeration areas (EAs) were chosen on a basis of probability proportional to size. Twenty-five households were sampled per EA using systematic random sampling as per the second stage. All eligible persons were all women aged 15 49 years, and were usual residents of the selected households, or had stayed in the selected households the night before the survey. This was an analysis of women who were on subsampled households and who had completed PHQ-9 depression module. The weighted analytic sample was the final sample of 3, 297 ever-married women aged 15- 49 [**Figure 1**].

[**Insert Figure 1 Here**]

Outcome variable

The main outcome variable Patient Health Questionnaire-9 (PHQ-9) was used to evaluate the depressive symptoms. A PHQ-9 score of 10 and above was considered a moderate-severe depressive symptom in women whereas a score below 10 was considered no-depression. All regression analysis used this binary outcome. Internal consistency of the PHQ-9 in this sample was assessed using Cronbach's alpha ($\alpha = 0.82$), indicating good reliability. (M. S. Miah & Ullah, 2025; Westby et al., 2021).

Exposure Variable Decision-Making Autonomy

The main exposure variable was decision-making autonomy which was measured by a composite index of empowerment based on five domains of household and reproductive choices:

1. Own health care.
2. Large household purchases.
3. Use of household money.
4. Visits to relatives or family.
5. Family planning decisions.

Women were regarded autonomous in each domain if they reported making the decision alone or jointly with their partner. Each empowered domain contributed 1 point, resulting in a composite score from 0 to 5. Then, this study considered, no autonomy: composite score ≤ 2 and yes autonomy: composite score ≥ 3. This binary variable (Decision-Making Autonomy) was used as the main exposure in all analyses examining its association with depressive symptoms. This method corresponds with the past research conducted by the DHS on the level of empowerment of women and represents the meaningful involvement in the household decision-making process. his binary measure captures the overall level of meaningful involvement in household and reproductive decisions, providing a clear indicator of women's autonomy for analysis (Ameyaw et al., 2016; Anfaara et al., 2024; Antabe et al., 2025; Kamiya, 2011; M. S. U. M. O. Miah, 2025).

Covariates

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In this study, women's sociodemographic characteristics were categorized in line with standard DHS methods. Women's age was categorized into seven groups (15–19, 20–24, 25–29, 30–34, 35–39, 40–44, 45–49 years), while age at first birth was grouped as early (10–17 years), normal (18–24 years), and late (≥25 years). Age at first sexual intercourse was categorized as <18 years or ≥18 years, and age at cohabitation was classified into young (<18 years), mild (18–20 years), and older (≥21 years). Women's education was categorized into below primary, secondary incomplete, and secondary or higher, and maternal employment was categorized as currently working or not working. Adequate antenatal care (ANC) visits were dichotomized as <4 or ≥4 visits. Reproductive characteristics included number of children (none, 1–2, 3+), birth size (small, average or large), preterm birth (term ≥9 months, preterm <9 months), current pregnancy status (yes/no), pregnancy multiplicity (single/multiple), pressure to be pregnant (yes/no), and history of pregnancy loss (none, one loss, two or more losses). Residing status was classified as residing with or not with husband/partner, current abstinence as yes/no, number of unions as once or more than once, and recent menstruation as yes/no. IPV justification (does not justify/justifies IPV). Household environment and socioeconomic factors included place of residence (urban/rural), administrative province (Butha-Buthe, Leribe, Berea, Maseru, Mafeteng, Mohale's Hoek, Quthing, Qacha's Nek, Mokhotlong, Thaba-Tseka), religion (Roman Catholic/Other), wealth status (poor, middle, rich), household size (1–3, 4+ members), household head sex (male/female), household assets (ownership of motorbike or bicycle; yes/no), media access (access to TV, radio, newspaper, or mobile; yes/no), and household construction materials (based on roof materials; improved/unimproved). Source of drinking water and sanitation facility were also categorized as improved or unimproved. Internet use in the last 12 months was coded as yes/no. All covariate categories were harmonized with previous DHS-based studies to ensure consistency in reporting and comparability across surveys and analyses (Esteban-Gonzalo et al., 2026; M. S. Miah et al., 2026; Tamanna et al., 2025; Tamanna & Mahmud, 2025; Zoha et al., 2025).

Data quality and management

The data were entered electronically (CSPro) and checked automatically in terms of range and consistency. Supervision of data on a daily basis, transfer in safe data and central data cleaning guaranteed high data quality. Additional checking and cleaning of secondary data were performed~~done~~ before final analysis.

Statistical analysis

All the analyses factored in the complicated survey design with sampling weights, clustering, and stratification in order to make it representative of the nation. An analysis of complete-cases was applied.

Code Availability

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The cleaned dataset, analysis scripts, and related resources supporting the findings of this study are available at GitHub:

https://github.com/muhammadsalek/Rural-Urban-Inequalities-in-Depression-among-Ever-Married-Women-in-Lesotho. The analyses were conducted following standard public health and epidemiological approaches and are consistent with methods applied in previous literature (Celentano ScD MHS & Szklo MPH DrPH, 2019).

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Descriptive, and bivariate analysis

The characteristics of the participants were summarized using descriptive statistics. Chi-square tests were used to test bivariate relationship between depressive symptoms and explanatory variables. Multivariate modeling was considered on variables having p less than 0.20 or good conceptual relevance (Celentano ScD MHS & Szklo MPH DrPH, 2019). Spatial analysis to depict spatial difference in the prevalence of depression and autonomy of women in decision-making at the province level, choropleth maps were shaped.

Multivariable modeling

Logistic regression was used to assess associations between depressive symptoms and decision-making autonomy. Models were built sequentially using a forward stepwise approach, informed by both statistical significance and prior literature:

- **Model 0:** Maternal age and decision-making autonomy variable just.
- **Model 1:** Sociodemographic factors (education, religion, residence).
- **Model 2:** Reproductive factors (age at first birth, cohabitation, parity, multiple pregnancy, pregnancy loss).
- **Model 3:** Contextual factors (urban/rural residence)

- **Model 4 (Final model):** Household and environmental factors (toilet facility, water source, household size, household head sex, wealth, media and internet access, household assets and materials).

The final model was nominated based on conceptual relevance and improved model fit (AIC = 1648.22; BIC = 1886.15; mean VIF = 2.2).

Final Model Adjusted Covariates

This set of confounders were adjusted in the final balanced model including women age, women education, maternal employment, religion, administrative division, age at first birth, age at cohabitation, number of children, pregnancy status, pregnancy loss, place of residence, toilet facilities, water access, household head sex, household size category, household wealth status, media access, internet use, household assets, and household materials.

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Model diagnostics

Multicollinearity was assessed using variance inflation factors (VIF). Model fit was assessed using the Hosmer–Lemeshow test, which showed good fit ($\chi^2 = 5.85$, $p = 0.6642$). Model discrimination was assessed using the area under the ROC curve (AUC = 0.6781), demonstrating moderate predictive ability. Results are reported as adjusted odds ratios (aORs) with 95% confidence intervals, with statistical significance set at $p < 0.05$.

Sensitivity Analysis

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In this study, no additional sensitivity analyses were conducted for the PHQ-9 scores, as we adhered to the standard DHS scoring protocol for defining moderate-severe depressive symptoms (DHS Program, 2023).

Ethical considerations

The LDHS protocol acknowledged ethical approval from relevant ethics committees in Lesotho and the ICF Institutional Review Board. Written informed consent was obtained from all participants. This study used anonymized, publicly available DHS data with permission from the DHS Program. However, details available at at https://dhsprogram.com/Methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm .

Results

The prevalence of depression was 7.4% (95% CI: 6.3-8.7). On place of residence, just over half (54.98%) respondents reported the rural (95% CI: 51.47-58.45) and 45.02% reported the urban area (95% CI: 41.55-48.53). The small overlap between the confidence limits implies that more rural participants are represented in the study population though, to establish the statistical significance of the difference, this would have to be tested. The confidence intervals of the two categories of depression and non-depression are complementary as expected, and it proves that depression is a minor yet non-trivial burden to the population health [**Figure 2,Figure 3**].

[**Insert Figure 2 and Figure 3 Here**]

The prevalence of the rural population in the study population was 54.98% (95% CI: 51.47-58.45) and that of the urban population was 45.02% (95% CI: 41.55-48.53). The fact that the CIs are mainly non-overlapping implies that the percentage of rural members might be higher than that of urban people, yet it is necessary to test whether the difference is significant. The depression prevalence was 7.40% (95% CI: 6.26-8.74) meaning it is possible to suggest that depression can be experienced by 6-9% of the target population. On the other hand, 92.60% respondents were not depressed (95% CI: 91.26–93.74), which is an accurate estimate since the percentage in this group was large. All in all, these results suggest that the prevalence of depression is lower than in the majority of the population, and rural inhabitants are a slightly higher proportion of the study sample [**Figure 4**].

[**Insert Figure 4 Here**]

On the whole, depressive symptoms were observed only in a few prevalences of women. Particularly, higher among women in the age groups of 15-19 years, 25-29 years, and 35-39 years. On the other hand, early reproductive such as early fertility, early onset of sexual activity, and early cohabitation were associated with relatively high levels of depressive symptoms. The variations due to socioeconomic status were relatively low, with women having lower education levels and being unemployed displaying relatively high prevalence. In relation to health determinants, limited variation was noted apart from women who had relatively fewer antenatal visits and women with high parity having a higher prevalence of depressive symptoms. Women who lacked decision-making power and those from urban centers had relatively high depression levels. The variations by district were also significant

~~All age groups showed depressive symptoms, with a relative overrepresentation of women aged between 15-19, 25-29, and 35-39 years, although confidence intervals were overlapping, and therefore, there was no evident age gradient at the descriptive level. Women who had early life experiences, such as younger age at first birth, young age at sexual debut (< 18 years) and young age at cohabitation were found to have more depressive symptoms, which is consistently higher than their counterparts, indicating greater vulnerability of women who were exposed to early reproductive and marriage events. Equally, women who reported pressure to become pregnant, a history of pregnancy loss, and a large household size had a higher prevalence of depressive symptoms, which indicates the possibility of the involvement of psychosocial and reproductive~~

~~stressors. Patterns of socioeconomic mixture were heterogeneous. Women with lower education levels, poor households, and women who were not employed at all had slightly higher rates of depressive symptoms, although similarities in the confidence interval indicate slight variations. It is worth noting that women living in urban regions showed greater incidences of depressive symptoms as compared to those in rural regions, which showed potential influences of the context or the environment. There was also little variance in relation to health-related factors: depressive symptoms were slightly more in women having received insufficient antenatal care, and greater parity and multiple pregnancies, whereas preterm birth and current pregnancy status were not significantly different. The use of media and internet exposure did not exhibit high levels of descriptive differences. Depressive symptoms were heterogeneous over the districts such that Maseru, Leribe, and Berea showed a higher prevalence and therefore, there was spatial heterogeneity which needs multivariate and spatial analysis. On the whole, the descriptive results reveal that depressive symptoms are concentrated around transitional experiences in early life, reproductive stressors, and urban living, which explains the necessity to implement context-dependent and equity-based mental health programs. Independent associations have to be determined by formal inferential analyses~~ [**Table 1**].

[**Insert Table 1 Here**]

When controlling the odds of depression against a comprehensive set of sociodemographic, reproductive, household, and contextual confounders, women with an autonomy of decision made were revealed to have much lower odds of depression at the national level (AOR = 0.63; 95% CI: 0.39-0.98, p=0.048). Additionally, there was a significant rural-urban heterogeneity indicated by stratified analyses. Decision-making autonomy in rural areas was significantly associated to lower chances of having depressive symptoms (AOR = 0.44; 95% CI: 0.27-0.~~19~~74, p=0.002) only [**Table 2**].

[**Insert Table 2 Here**]

An AUC of 0.68 implies that, in approximately 68% of randomly selected case–non-case pairs, the model assigns a higher predicted probability to the woman with the outcome than to the woman without the outcome [**Figure 5**].

[**Insert Figure 5 Here**]

Lower levels of autonomy were concentrated in the southern and south-western provinces (Quthing, Qacha’s Nek, Mohale’s Hoek, and Mafeteng), which represent high-burden areas of limited decision-making power. In contrast, the central and northern provinces, including Thaba-Tseka and Maseru, exhibited relatively higher autonomy, although overall empowerment levels remained modest. These geographic patterns highlight meaningful regional variation in female autonomy and identify priority areas for context-specific interventions~~The lower levels of autonomy were centralized towards the southern and south-western provinces, the provinces of Quthing, Qacha Nek, Mohale Hoek and Mafeteng, which indicated high burden areas of low autonomy. That is not the case in central and northern provinces, including Thaba Tseka and Maseru, which were relatively more autonomous, which are lower-burden provinces, but autonomy was nevertheless poor in general. These geographical patterns connect the existence of definite geographic inequalities in female empowerment and define the priority areas where the most context-specific intervention might be required~~ [**Figure 7**].

[**Insert Figure 7 Here**]

The central-northern districts, especially the Maseru, Berea, Leribe, and Butha-Buthe, proved to have a higher prevalence of depression, which resulted in the identification of these districts as high-burden areas. Conversely, it was showed to be lower in the southern and south-eastern districts, which comprise of Quthing, Qachas Nek, Mafeteng, and some parts of Hoek of Mohale which are relatively low-burden areas. These geographic inequalities reveal that there exist significant subnational differences in mental health burden and that mental health planning and resource allocation needs to be done at the district level with a focus on more burdensome central-northern districts [**Figure 6**].

[**Insert Figure 6 Here**]

Discussion

Using nationally representative LDHS 2023–24 data, this study found that ~~7.4% of~~ women in Lesotho experienced ~~moderate-to-severe~~moderate-severe depressive symptoms, confirming a non-trivial mental health burden. ~~Women's decision-making autonomy was significantly associated with lower odds of depression at the national level (AOR = 0.63; 95% CI: 0.39–0.98), with a strong protective effect in rural areas (AOR = 0.44; 95% CI: 0.27–0.74), while no significant association was observed among urban women.~~ Spatial analysis revealed substantial geographic inequalities: central–northern districts, including Maseru, Berea, Leribe, and Butha-Buthe, had the highest prevalence of depression, whereas southern and south-western districts (Quthing, Qacha's Nek, Mohale's Hoek, Mafeteng) exhibited the lowest autonomy and comparatively lower depression prevalence. These findings highlight the importance of women's decision-making autonomy for mental health, particularly in rural areas, and underscore spatial disparities that warrant targeted interventions. These findings are interpreted within the social determinants of mental health framework, highlighting those structural inequities, reproductive stressors, and decision-making autonomy shape women's vulnerability to depressive symptoms in Lesotho.

Previous studies have shown that depressive symptoms are a growing concern in Southern Africa, particularly among women facing social and economic vulnerabilities (Melkam et al., 2025; Mokwena & Makhozonke, 2024). Our study adds new evidence by linking decision-making autonomy directly to depressive symptoms at both national and spatial levels, highlighting rural–urban and geographic disparities (Noor, Siddique, Yeasar, et al., 2025).

According to our findings, the sample proportion of rural residents was a little bit higher (55%), but the rates of depressive symptoms were slightly higher in women in the urban group. The observed rural–urban differences may reflect the distinct social and economic environments in Lesotho. Although rural women are not easily accessible to professional mental health care providers, their social and family networks provide them with more advantages in terms of making decisions and dealing with problems, increasing the protective value of autonomy (Acharya et al., 2010b; Ameyaw et al., 2016). On the other hand, even though urban women have more resources at their disposal, they are more prone to economic hardship, social isolation, and urban stress, all of which reduce the protective nature of autonomy on depression (Antabe et al., 2025). This is contrary to other LMIC studies where urban women can usually report improved mental health outcomes because of increased socioeconomic resources (Asmare et al., 2026). However, in Lesotho, urban life can be accompanied by a greater exposure to stressors, including economic, social isolation or urban poverty, which may at least partly be responsible for the observed urban vulnerability (Mokwena & Makhozonke, 2024). It was determined that events and pressures in early life including the earlier age of first birth, earlier sexual initiation, earlier cohabitation, and pregnancy loss were connected with increased depressive symptomatology (Moeti et al., 2022;

Russell et al., 2021). These results are in line with the previous studies that associate reproductive stressors and early life transitions to mental health vulnerability in the long term as a woman (Bushu et al., 2025). It is important to note that the association~~correlation~~ between the depressive symptoms and pregnancy loss highlights the urgent necessity of providing women with reproductive difficulties with a specific kind of psychosocial assistance (Boakye et al., 2025).

Taken together, these findings suggest that empowering women in decision-making and addressing early-life reproductive stressors are critical for reducing depressive symptoms. The implications for policy include designing context-specific, equity-oriented interventions that combine reproductive and mental health support, with particular attention to rural areas and high-burden districts.

Less education and unemployment and poor household wealth were also associated with it but with slightly high prevalence of depression, though it was within the same confidence interval (Quenby et al., 2021). This is in line with the current social determinants of mental health framework that identifies structural inequities as a way of aggravating the risk factors on the individual level (Alkema et al., 2016). In the national level, autonomy in making decisions showed lower odds with depression ~~(AOR = 0.63; 95% CI: 0.39-0.98)~~, and this association was even stronger in rural settings ~~(AOR = 0.44; 95% CI: 0.27-0.97)~~. The results are consistent with other studies on Southern Africa that showed women empowerment and autonomy to be protective against depressive symptoms (Miller et al., 2016). The predictive performance of the multivariable model was modest around (AUC = 0.68), indicating fair discrimination. While the model performs better than chance, it does not achieve a level of accuracy sufficient for individual-level prediction. This is consistent with prior population-based mental health studies, where complex psychosocial outcomes are influenced by unmeasured factors such as interpersonal relationships, trauma, and social norms (Ostovar et al., 2025). As another finding of our spatial analyses, central-northern districts showed more depression prevalence whereas southern and south-western provinces (Quthing, Qacha Nek, Mafeteng, Mohale Hoek) showed more autonomy and less depression (DHS Program, 2023; Mokwena & Makhozonke, 2024). These trends can be aligned with the hypothesis that geographic disparities in enabling and psychosocial stress are associated with unequal mental health workloads in districts (Thibaut & ELNahas, 2023). Interestingly, health-related variables like antenatal care, parity and multiple pregnancies were found to have minimum dependence on depression whereas media and internet exposure did not seem to have any effects on the symptom patterns (Flor et al., 2026). This is contrary to other LMICs where utilization of health services tends to mediate the mental health outcomes implying that the context of social and other determinants of mental health may play a higher role in the Lesotho context(M. S. Miah & Ullah, 2025; Noor, Siddique, Das, et al., 2025; Pandey et al., 2024). Altogether, these results demonstrate the need of context-specific, equity-oriented interventions, which combine both reproductive and mental health support (Zhang et al., 2019). The observed spatial variations, with higher depression prevalence in central–northern districts and lower prevalence in southern and south-western provinces, are likely due to geographic differences in women's autonomy, socioeconomic resources, psychosocial stressors, and urban-rural contextual factors. These findings revealed the need for geographically targeted mental health interventions (Abedin & Arunachalam, 2020). The identified geographic and sociodemographic differences also suggest that national mental health policy must focus on central-northern districts and women with early-life reproductive stressors, as well as encourage autonomy in decision-making as a protective factor (Llop-Gironés et al., 2019). The combination of spatial and individual-level analysis in this research offers practical piece of evidence in order to guide the allocation of resources and specific mental health programming in Lesotho (Dangare et al., 2026).

Overall, this study demonstrates what was previously known, adds novel insights into the protective role of autonomy and geographic disparities, and underscores the need for tailored, empowerment-focused mental health interventions in Lesotho.

Policy Implication and Recommendations

The findings indicate that mental health services in Lesotho need to be provided in context-specific manner. Policies should: Empower rural women who have a low autonomy, improve empowerment initiatives, reproductive decision-making support, and intervention aimed at psychosocial. Increase central-northern districts with high burden of mental health in need of expanded services, community awareness, and primary care integration. Integrate the empowerment strategies into the wider maternal and reproductive health programs, recognizing the differing impacts between the rural and the urban areas.

Strengths and Limitations

The strengths are the utilization of national representative data, the measurement of depression based on PHQ-9, and the incorporation of the spatial analysis that can be used to demonstrate geographic inequities. This study has several limitations, such as its cross-sectional design, which excludes the possibility of causal inference, and use of self-reported measures of autonomy and depression, which is prone to reporting bias. In spite of these shortcomings, the article presents strong evidence on how the issues of empowerment, reproductive factors and mental health intersect in Lesotho. Additionally, the PHQ-9 relies on self-reported symptoms and may not fully capture clinical depression, and its sensitivity and specificity can vary across cultural and demographic contexts. Future studies may consider alternative PHQ-9 cut-offs or item-level sensitivity checks to further assess robustness.

Conclusions

Despite these limitations, the study revealed strong evidence on the association between decision making autonomy and depressive symptoms with rural-urban disparities in Lesotho. These results indicate that the autonomy of women can be a protective factor against depression in rural areas, where the structural and social limitations to the agency of women may be higher. On the whole, the findings reveal context-specific mental health gains of women empowerment and the need to incorporate in rural maternal and mental health interventions autonomy-enhancing strategies. The results indicate that there exist high rural-urban inequalities, geographic inequalities, central-northern districts had prevalence of depression and lowest autonomy was observed in southern-southwestern districts. These findings highlight the importance of context-sensitive, empowerment-based mental health services, especially the rural women and high-burden districts. The combination of women empowerment strategies with reproductive health and mental health initiatives might help to decrease the symptoms of depression and help to ensure equality in mental health outcomes nationwide.

Declarations

Data Availability

The data used in this study are publicly available from the Demographic and Health Surveys (DHS) Program and can be requested at https://www.dhsprogram.com/data.

Assistance of Generative AI Tools

AI-assisted tools including Grammarly, QuillBot, and ChatGPT were used solely for language editing and grammar improvements. AI did not contribute to data analysis, interpretation, or intellectual content. The authors verified the manuscript for originality and ethical compliance.

Consent to participant

Not applicable.

Consent for Publication

Not applicable. This study used de-identified, publicly available data and does not include any identifiable information.

Conflict of Interest

The authors declare no competing interests.

CRediT authorship contribution statement

Md Salek Miah: Solely responsible for Conceptualization, Data Curation, Methodology, Resources, Project Administration, Formal Analysis, and all stages of Writing – Original Draft, Writing – Review & Editing of the Final Manuscript.

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Maria Bintey Kabir: Assisting to Handle of reviewers' comments and Editing of the Draft Manuscript.

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~~**Md Salek Miah:** Conceptualization, Data Curation, Methodology, Formal Analysis, Writing – Original Draft and Final Manuscript, Resources, Project Administration.~~

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~~Code Availability~~

~~All codes and analytical datasets used in this study are available from the first author upon request.~~

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References

Abedin, S., & Arunachalam, D. (2020). Maternal autonomy and high-risk pregnancy in Bangladesh: The mediating influences of childbearing practices and antenatal care. *BMC Pregnancy and Childbirth*, *20*(1). https://doi.org/10.1186/s12884-020-03260-9

Acharya, D. R., Bell, J. S., Simkhada, P., Van Teijlingen, E. R., & Regmi, P. R. (2010a). Open Access RESEARCH Women's autonomy in household decision-making: a demographic study in Nepal. In *Reproductive Health* (Vol. 7). http://www.reproductive-health-journal.com/content/7/1/15

Acharya, D. R., Bell, J. S., Simkhada, P., Van Teijlingen, E. R., & Regmi, P. R. (2010b). Women's autonomy in household decision-making: A demographic study in Nepal. *Reproductive Health*, *7*(1). https://doi.org/10.1186/1742-4755-7-15

Alkema, L., Chou, D., Hogan, D., Zhang, S., Moller, A. B., Gemmill, A., Fat, D. M., Boerma, T., Temmerman, M., Mathers, C., & Say, L. (2016). Global, regional, and national levels and trends in maternal mortality between 1990 and 2015, with scenario-based projections to 2030: A systematic analysis by the un Maternal Mortality Estimation Inter-Agency Group. *The Lancet*, *387*(10017), 462–474. https://doi.org/10.1016/S0140-6736(15)00838-7

Ameyaw, E. K., Tanle, A., Kissah-Korsah, K., & Amo-Adjei, J. (2016). Women's Health Decision-Making Autonomy and Skilled Birth Attendance in Ghana. *International Journal of Reproductive Medicine*, *2016*, 1–9. https://doi.org/10.1155/2016/6569514

Amouzou, A., Barros, A. J. D., Requejo, J., Faye, C., Akseer, N., Bendavid, E., Blumenberg, C., Borghi, J., El Baz, S., Federspiel, F., Ferreira, L. Z., Hazel, E., Heft-Neal, S., Hellwig, F., Liu, L., Maïga, A., Munos, M., Pitt, C., Shawar, Y. R., … Boerma, T. (2025). The 2025 report of the Lancet Countdown to 2030 for women's, children's, and adolescents' health: tracking progress on health and nutrition. *The Lancet*, *405*(10488), 1505–1554. https://doi.org/10.1016/S0140-6736(25)00151-5

Anfaara, F. W., Amoak, D., Kye, N. O., Sano, Y., & Antabe, R. (2024). Exploring the link between household structure and women's household decision-making autonomy in Mauritania. *Journal of Biosocial Science*. https://doi.org/10.1017/S0021932024000221

Antabe, R., Antabe, G., Sano, Y., & Pienaah, C. K. A. (2025). Women's household decision-making autonomy and mental health outcomes in Mozambique. *Global Mental Health*, *12*. https://doi.org/10.1017/gmh.2025.29

Asmare, A. A., Tegegne, A. S., & Belay, D. B. (2026). Unveiling hidden heterogeneity and inequalities in the continuum of care for reproductive, maternal, and child health services in sub-Saharan Africa: A multilevel latent class analysis approach. *Global Epidemiology*, *11*. https://doi.org/10.1016/j.gloepi.2025.100237

Bandara, N. A., Al-Anzi, S. M. F., Zhdanova, A., & Hirani, S. (2024). Women's Empowerment and Mental Health: A Scoping Review. *Women*, *4*(3), 277–289. https://doi.org/10.3390/women4030021

Boakye, S. K., Danquah, L., & Antwi, M. (2025). Maternal mental health improvement in ghana: role of green spaces exposure. *BMC Women's Health*, *25*(1). https://doi.org/10.1186/s12905-025-03886-x

Borges do Nascimento, I. J., Abdulazeem, H. M., Weerasekara, I., Marquez, J., Vasanthan, L. T., Deeken, G., Morgan, R., Tan, H. L., Yordi Aguirre, I., Østeengaard, L., Kularathne, I., Azzopardi-Muscat, N., van Kessel, R., Martinez, E. Z., Permanand, G., & Novillo-Ortiz, D. (2025). Transforming women's health, empowerment, and gender equality with digital health: evidence-based policy and practice. In *The Lancet Digital Health* (Vol. 7, Number 6). Elsevier Ltd. https://doi.org/10.1016/j.landig.2025.01.014

Bushu, G., Kambale, C., Kikwabantu, S., Benedite, F. B., Weber, W., Black, B. O., & Verschueren, K. J. C. (2025). Caesarean section Robson classification, complications and lessons learned in a rural hospital in Walikale, North Kivu, Democratic Republic of Congo: a cross-sectional study. *AJOG Global Reports*, 100586. https://doi.org/10.1016/j.xagr.2025.100586

Celentano ScD MHS, D. D., & Szklo MPH DrPH, M. (2019). *Unlock your eBook today*.

Cuschieri, S. (2019). The STROBE guidelines. In *Saudi Journal of Anaesthesia* (Vol. 13, Number 5, pp. S31–S34). Wolters Kluwer Medknow Publications. https://doi.org/10.4103/sja.SJA_543_18

Dangare, P., Cronje, P. J. R., Mashimbye, Z. E., Masanganise, J., Ntshidi, Z., Gokool, S., Naiken, V., Sawunyama, T., & Dzikiti, S. (2026). Modelling water - use and yield of selected irrigated subtropical crops using machine learning and hybrid models in north - eastern South Africa. *Agricultural Water Management*, *324*. https://doi.org/10.1016/j.agwat.2025.110113

DHS Program, T. (2023). *Lesotho Demographic and Health Survey*.

Esteban-Gonzalo, S., Khan, M. H. A., Hasan, M. M., & Díaz-Morales, J. F. (2026). Psychosocial predictors of women's mental health in Bangladesh: Evidence from BDHS-2022 data on fertility, empowerment, decision-making, child loss, and sex-affective behaviours. *Public Health*, *254*. https://doi.org/10.1016/j.puhe.2026.106222

Fernández, L. G., Yoon, G. H., Firima, E., Gupta, R., Sematle, M. P., Khomolishoele, M., Molulela, M., Bane, M., Tlahani, M., Lee, T. T., Chammartin, F., Gerber, F., Lejone, T., Ayakaka, I., Labhardt, N. D., Amstutz, A., & Belus, J. M. (2025). Prevalence of Mental Health and Substance use Problems and Awareness of Need for Services in Lesotho: Results from a Population-Based Survey. *International Journal of Mental Health and Addiction*, *23*(5), 3607–3626. https://doi.org/10.1007/s11469-024-01309-w

Flor, L. S., Spencer, C. N., Cagney, J., Gil, G. F., Aalruz, H., Abd ElHafeez, S., Abdelwahab, S. I., Abdoun, M., Abebe, M., Abebe, Y., Abedi, A., Abeldaño Zuñiga, R. A., Abie, A., Abiodun, O., Aboagye, R. G., Abreu, L. G., Abu Farha, R. K., Abubakar, B., Abuhammad, S., … Gakidou, E. (2026). Disease burden attributable to intimate partner violence against females and sexual violence against children in 204 countries and territories, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet*, *407*(10523), 31–52. https://doi.org/10.1016/S0140-6736(25)02503-6

Gebeyehu, N. A., Gelaw, K. A., Lake, E. A., Adela, G. A., Tegegne, K. D., & Shewangashaw, N. E. (2022). Women decision-making autonomy on maternal health service and associated factors in low- and middle-income countries: Systematic review and meta-analysis. In *Women's Health* (Vol. 18). SAGE Publications Ltd. https://doi.org/10.1177/17455057221122618

Heise, L., Greene, M. E., Opper, N., Stavropoulou, M., Harper, C., Nascimento, M., Zewdie, D., Darmstadt, G. L., Greene, M. E., Hawkes, S., Henry, S., Heymann, J., Klugman, J., Levine, R., Raj, A., & Rao Gupta, G. (2019). Gender inequality and restrictive gender norms: framing the challenges to health. In *The Lancet* (Vol. 393, Number 10189, pp. 2440–2454). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(19)30652-X

Herrman, H., Patel, V., Kieling, C., Berk, M., Buchweitz, C., Cuijpers, P., Furukawa, T. A., Kessler, R. C., Kohrt, B. A., Maj, M., McGorry, P., Reynolds, C. F., Weissman, M. M., Chibanda, D., Dowrick, C., Howard, L. M., Hoven, C. W., Knapp, M., Mayberg, H. S., … Wolpert, M. (2022). Time for united action on depression: a Lancet–World Psychiatric Association Commission. *The Lancet*, *399*(10328), 957–1022. https://doi.org/10.1016/S0140-6736(21)02141-3

Hindman, J., Zugai, J., Foster, K., & Raeburn, T. (2026). Self-Determination Theory as a Framework for Research and Design of Digital Applications for Nurses' Well-Being. In *Journal of Advanced Nursing* (Vol. 82, Number 1, pp. 897–906). John Wiley and Sons Inc. https://doi.org/10.1111/jan.16895

Hormazábal-Salgado, R., Whitehead, D., Osman, A. D., & Hills, D. (2024). Person-Centred Decision-Making in Mental Health: A Scoping Review. In *Issues in Mental Health Nursing* (Vol. 45, Number 3, pp. 294–310). Taylor and Francis Ltd. https://doi.org/10.1080/01612840.2023.2288181

Idris, I. B., Hamis, A. A., Bukhori, A. B. M., Hoong, D. C. C., Yusop, H., Shaharuddin, M. A. A., Fauzi, N. A. F. A., & Kandayah, T. (2023). Women's autonomy in healthcare decision making: a systematic review. *BMC Women's Health*, *23*(1). https://doi.org/10.1186/s12905-023-02792-4

Jejaw, M., Kelkay, J. M., Demissie, K. A., Hagos, A., Teshale, G., Baykemagn, N. D., Tadele, A., Tafere, T. Z., Aschalew, A. Y., & Yehuala, T. Z. (2026). Estimated Impact of Pressure to Conceive on Maternal Depression in Lesotho: A Quasi-experimental Propensity Score Matching using DHS Data. *Archives of Women's Mental Health*, *29*(2). https://doi.org/10.1007/s00737-026-01687-1

Jia, J., Snider, A., Collishaw, A., McNamara, P. E., & Tumusiime, E. (2025). Pathways linking WASH access and women's empowerment: Evidence from Zambia and Honduras. *Journal of Rural Studies*, *116*. https://doi.org/10.1016/j.jrurstud.2025.103602

Kamiya, Y. (2011). Women's autonomy and reproductive health care utilisation: Empirical evidence from Tajikistan. *Health Policy*, *102*(2–3), 304–313. https://doi.org/10.1016/j.healthpol.2011.04.001

Leight, J., Pedehombga, A., Ganaba, R., & Gelli, A. (2022a). Women's empowerment, maternal depression, and stress: Evidence from rural Burkina Faso. *SSM - Mental Health*, *2*. https://doi.org/10.1016/j.ssmmh.2022.100160

Leight, J., Pedehombga, A., Ganaba, R., & Gelli, A. (2022b). Women's empowerment, maternal depression, and stress: Evidence from rural Burkina Faso. *SSM - Mental Health*, *2*. https://doi.org/10.1016/j.ssmmh.2022.100160

Leseba, N., Vermaak, K., Makatjane, T., & Lebuso, M. (2025). A multilevel analysis of factors associated with stunting among children under five years in Lesotho: a study of the lesotho multiple cluster indicator survey 2018. *Journal of Health, Population and Nutrition*, *44*(1). https://doi.org/10.1186/s41043-025-00901-7

Llop-Gironés, A., Julià, M., Chicumbe, S., Dulá, J., Odallah, A. A. P., Alvarez, F., Zahinos, I., Mazive, E., & Benach, J. (2019). Inequalities in the access to and quality of healthcare in Mozambique: Evidence from the household budget survey. *International Journal for Quality in Health Care*, *31*(8), 577–582. https://doi.org/10.1093/intqhc/mzy218

Melkam, M., Fente, B. M., Negussie, Y. M., Asmare, Z. A., Asebe, H. A., Seifu, B. L., Gebrehana, A. K., Tsega, S. S., Bezie, M. M., & Asnake, A. A. (2025). Postpartum depression and associated factors among childbearing women from the recent Demographic and Health Survey data of Mozambique: Multilevel analysis. *Heliyon*, *11*(1). https://doi.org/10.1016/j.heliyon.2024.e41106

Miah, M. S., Lamia, F. F., Begum, Mst. L., Shied, M., & Akhter, R. (2026). Household Water and Handwashing Facilities and Early Childhood Education Participation in Nepal: Evidence From a Nationally Representative Cross-Sectional Survey. *Health Science Reports*, *9*(4). https://doi.org/10.1002/hsr2.72254

Miah, M. S. U. M. O. (2025). *Associations of women's empowerment and intimate partner violence with mental health among ever-married women in Bangladesh*.

Miah, M. S., & Ullah, M. O. (2025). Associations of internet use and pregnancy loss with depression and anxiety among women in Bangladesh: evidence from the 2022 BDHS. *BMC Women's Health*. https://doi.org/10.1186/s12905-025-04166-4

Miller, S., Abalos, E., Chamillard, M., Ciapponi, A., Colaci, D., Comandé, D., Diaz, V., Geller, S., Hanson, C., Langer, A., Manuelli, V., Millar, K., Morhason-Bello, I., Castro, C. P., Pileggi, V. N., Robinson, N., Skaer, M., Souza, J. P., Vogel, J. P., & Althabe, F. (2016). Beyond too little, too late and too much, too soon: a pathway towards evidence-based, respectful maternity care worldwide. In *The Lancet* (Vol. 388, Number 10056, pp. 2176–2192). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(16)31472-6

Moeti, M., Gao, G. F., & Herrman, H. (2022). Global pandemic perspectives: public health, mental health, and lessons for the future. In *The Lancet* (Vol. 400, Number 10353, pp. e3–e7). Elsevier B.V. https://doi.org/10.1016/S0140-6736(22)01328-9

Mokwena, K. E., & Makhozonke, N. (2024). The prevalence of postnatal depression and associated factors among clinic attendees in Maseru, Lesotho. *International Journal of Innovative Research and Scientific Studies*, *7*(2), 398–408. https://doi.org/10.53894/ijirss.v7i2.2644

Mustafiz, R. B., Chowdhury, S. S. A., Kundu, S., Munaf, N. B., & Sharif, A. Bin. (2025). Determinants of healthcare decision-making autonomy among Bangladeshi women: mixed-effect logistic regression analysis. *BMC Women's Health*, *25*(1). https://doi.org/10.1186/s12905-025-03666-7

Myint, W. W., Yusuf, A., Samman, E., Nguyen, A., Mendez, S. E., & Clark, H. R. (2025a). Factors associated with severity of depression among women intimate partner violence survivors in Lesotho: A cross-sectional study. *Women's Health*, *21*. https://doi.org/10.1177/17455057251387843

Myint, W. W., Yusuf, A., Samman, E., Nguyen, A., Mendez, S. E., & Clark, H. R. (2025b). Factors associated with severity of depression among women intimate partner violence survivors in Lesotho: A cross-sectional study. *Women's Health*, *21*. https://doi.org/10.1177/17455057251387843

Nabaggala, M. S., Reddy, T., & Manda, S. (2021). Effects of rural–urban residence and education on intimate partner violence among women in Sub-Saharan Africa: a meta-analysis of health survey data. *BMC Women's Health*, *21*(1). https://doi.org/10.1186/s12905-021-01286-5

Noor, S. T. A., Siddique, S., Das, O., Yeasar, S., & Islam, R. B. (2025). Exploring mental health disparities in Mozambique: Depression and anxiety symptoms among reproductive-aged women using data from Mozambique Demographic and Health Survey 2022–23. *Global Epidemiology*, *10*. https://doi.org/10.1016/j.gloepi.2025.100223

Noor, S. T. A., Siddique, S., Yeasar, S., Das, O., Tanvir, S.-U.-A., & Islam, R. B. (2025). *Prevalence, Determinants, and Care-Seeking Behaviour for Major Depressive Disorder in the Lesotho Population: A Multilevel Analysis from the Lesotho Demographic and Health Survey 2023-24*. https://doi.org/10.21203/rs.3.rs-6945317/v1

Okyere, J., Ayebeng, C., & Dickson, K. S. (2025a). Sex differences in help-seeking behavior for depression in Lesotho: findings from a national survey. *BMC Psychiatry*, *25*(1). https://doi.org/10.1186/s12888-025-06749-9

Okyere, J., Ayebeng, C., & Dickson, K. S. (2025b). Sex differences in help-seeking behavior for depression in Lesotho: findings from a national survey. *BMC Psychiatry*, *25*(1). https://doi.org/10.1186/s12888-025-06749-9

Oram, S., Fisher, H. L., Minnis, H., Seedat, S., Walby, S., Hegarty, K., Rouf, K., Angénieux, C., Callard, F., Chandra, P. S., Fazel, S., Garcia-Moreno, C., Henderson, M., Howarth, E., Macmillan, H. L., Murray, L. K., Othman, S., Robotham, D., Rondon, M. B., … Howard, L. M. (2022). *The Lancet Psychiatry Commissions The Lancet Psychiatry Commission on intimate partner violence and mental health: advancing mental health services, research, and policy* (Vol. 9). www.thelancet.com/psychiatry

Ostovar, A., Davari, D. D., & Dzikuć, M. (2025). Determinants of Design with Multilayer Perceptron Neural Networks: A Comparison with Logistic Regression. *Sustainability (Switzerland)*, *17*(6). https://doi.org/10.3390/su17062611

Pandey, A. R., Adhikari, B., Bista, B., Lamichhane, B., Joshi, D., Saugat Pratap, K. C., Sharma, S., & Baral, S. (2024). Prevalence, determinants and care-seeking behaviour for anxiety and depression in Nepalese population: a secondary analysis of data from Nepal Demographic and Health Survey 2022. *BMJ Open*, *14*(8). https://doi.org/10.1136/bmjopen-2023-078582

Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., Chisholm, D., Collins, P. Y., Cooper, J. L., Eaton, J., Herrman, H., Herzallah, M. M., Huang, Y., Jordans, M. J. D., Kleinman, A., Medina-Mora, M. E., Morgan, E., Niaz, U., Omigbodun, O., … UnÜtzer, Jü. (2018a). The Lancet Commission on global mental health and sustainable development. In *The Lancet* (Vol. 392, Number 10157, pp. 1553–1598). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(18)31612-X

Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., Chisholm, D., Collins, P. Y., Cooper, J. L., Eaton, J., Herrman, H., Herzallah, M. M., Huang, Y., Jordans, M. J. D., Kleinman, A., Medina-Mora, M. E., Morgan, E., Niaz, U., Omigbodun, O., … UnÜtzer, Jü. (2018b). The Lancet Commission on global mental health and sustainable development. In *The Lancet* (Vol. 392, Number 10157, pp. 1553–1598). Lancet Publishing Group. https://doi.org/10.1016/S0140-6736(18)31612-X

Quenby, S., Gallos, I. D., Dhillon-Smith, R. K., Podesek, M., Stephenson, M. D., Fisher, J., Brosens, J. J., Brewin, J., Ramhorst, R., Lucas, E. S., McCoy, R. C., Anderson, R., Daher, S., Regan, L., Al-Memar, M., Bourne, T., MacIntyre, D. A., Rai, R., Christiansen, O. B., … Coomarasamy, A. (2021). Miscarriage matters: the epidemiological, physical, psychological, and economic costs of early pregnancy loss. In *The Lancet* (Vol. 397, Number 10285, pp. 1658–1667). Elsevier B.V. https://doi.org/10.1016/S0140-6736(21)00682-6

Rahman, T., Gasbarro, D., Alam, K., & Alam, K. (2024). Rural–urban disparities in household catastrophic health expenditure in Bangladesh: a multivariate decomposition analysis. *International Journal for Equity in Health*, *23*(1). https://doi.org/10.1186/s12939-024-02125-3

Raz, S., Rahman, M. S., Aktar, B., & Rashid, S. F. (2025). Women's household decision-making autonomy in low economic settlements in Bangladesh: Evidence from a cross-sectional study, 2022? *Women's Studies International Forum*, *113*. https://doi.org/10.1016/j.wsif.2025.103181

Russell, L. B., Sobanjo-ter Meulen, A., & Toscano, C. M. (2021). Evaluating the cost-effectiveness of maternal pertussis immunization in low- and middle-income countries: A review of lessons learnt. In *Vaccine* (Vol. 39, Number 1, pp. 121–124). Elsevier Ltd. https://doi.org/10.1016/j.vaccine.2020.10.054

Tamanna, T., & Mahmud, S. (2025). Applying machine learning to predict quality ANC determinants in Bangladesh: a BDHS-2022 cross-sectional study. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-21411-w

Tamanna, T., Mahmud, S., Salma, N., Hossain, M. M., & Karim, M. R. (2025). Identifying determinants of malnutrition in under-five children in Bangladesh: insights from the BDHS-2022 cross-sectional study. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-99288-y

The Lancet. (2025). Empowering women during childbirth. In *The Lancet* (Vol. 406, Number 10498, p. 1). Elsevier B.V. https://doi.org/10.1016/S0140-6736(25)01382-0

Thibaut, F., & ELNahas, G. (2023). Women's Mental Health and Lessons Learnt from the COVID-19 Pandemic. In *Psychiatric Clinics of North America* (Vol. 46, Number 3, pp. 415–426). W.B. Saunders. https://doi.org/10.1016/j.psc.2023.04.001

Viens, L. J., Clouston, S., & Messina, C. R. (2016). Women's autonomy and cervical cancer screening in the Lesotho Demographic and Health Survey 2009. *Social Science and Medicine*, *150*, 23–30. https://doi.org/10.1016/j.socscimed.2015.12.009

Westby, C. L., Erlandsen, A. R., Nilsen, S. A., Visted, E., & Thimm, J. C. (2021). Depression, anxiety, PTSD, and OCD after stillbirth: a systematic review. *BMC Pregnancy and Childbirth*, *21*(1). https://doi.org/10.1186/s12884-021-04254-x

Zhang, Y., Augusto, O., Ásbjörnsdóttir, K., Akullian, A., Cumbe, V., Rao, D., Gimbel, S., Sherr, K., Wagenaar, B. H., Manuel, J. L., Hicks, L., Mahumane, A., Pfeiffer, J., Gloyd, S., Cuembelo, F., Nhumba, M., Lequechane, J., Covele, A., Napua, M., … Michel, C. (2019). Geographic distribution and determinants of mental health stigma in central Mozambique. *Social Psychiatry and Psychiatric Epidemiology*, *54*(11), 1391–1410. https://doi.org/10.1007/s00127-019-01708-8

Zoha, S., Alam, S., Sifat, I. K., Sultana, N., & Kibria, M. K. (2025). Identifying determinants and predicting cesarean section delivery among Bangladeshi women using machine learning: Insight from BDHS 2022 Data. *PLOS Global Public Health*, *5*(11 November). https://doi.org/10.1371/journal.pgph.0005494

List of All Figures

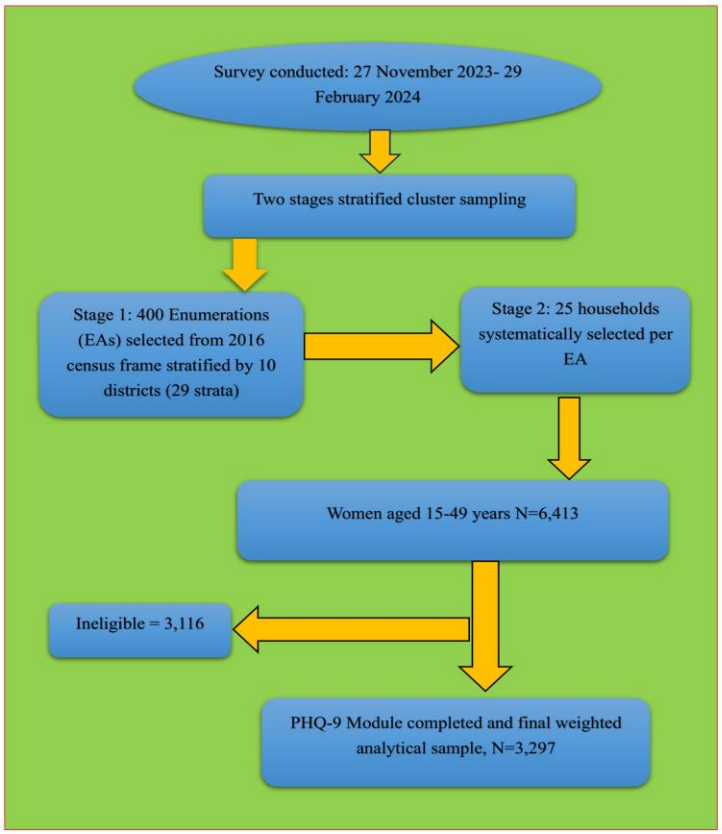

Figure 1: The Study Flow

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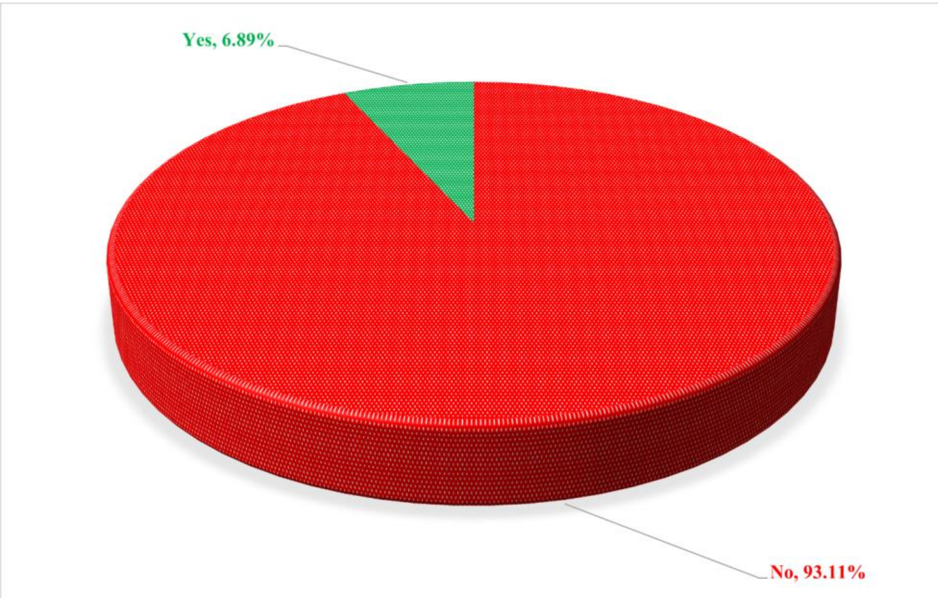

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Figure 2 : Prevalences of Depression Symptoms

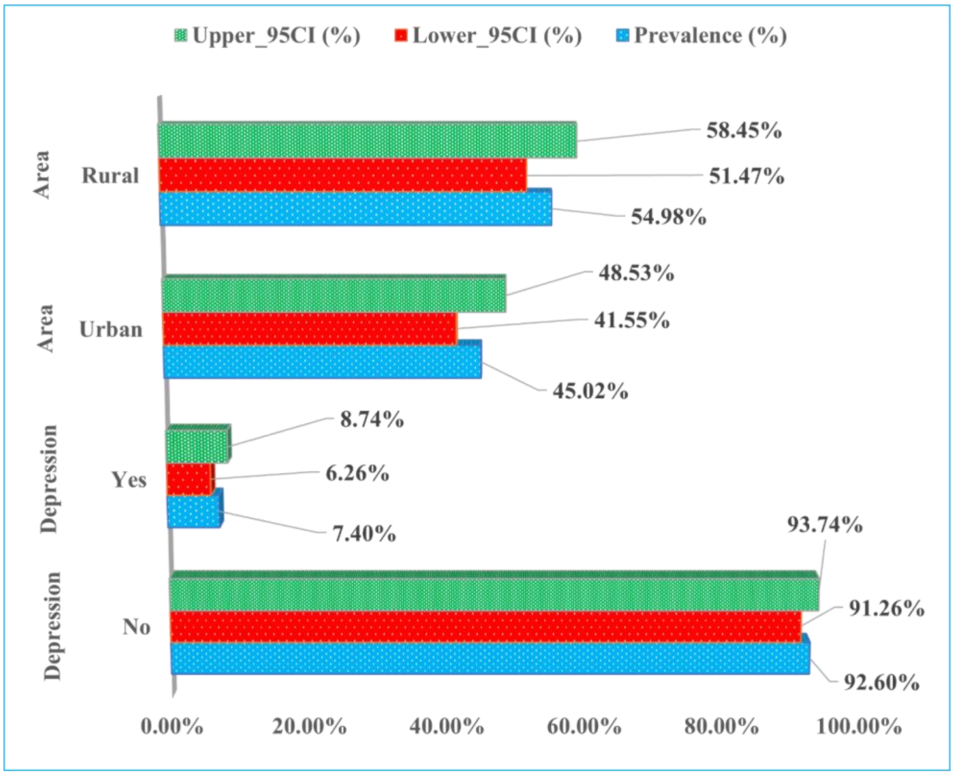

Figure 3: Rural-Urban Inequalities of Depression Prevalence

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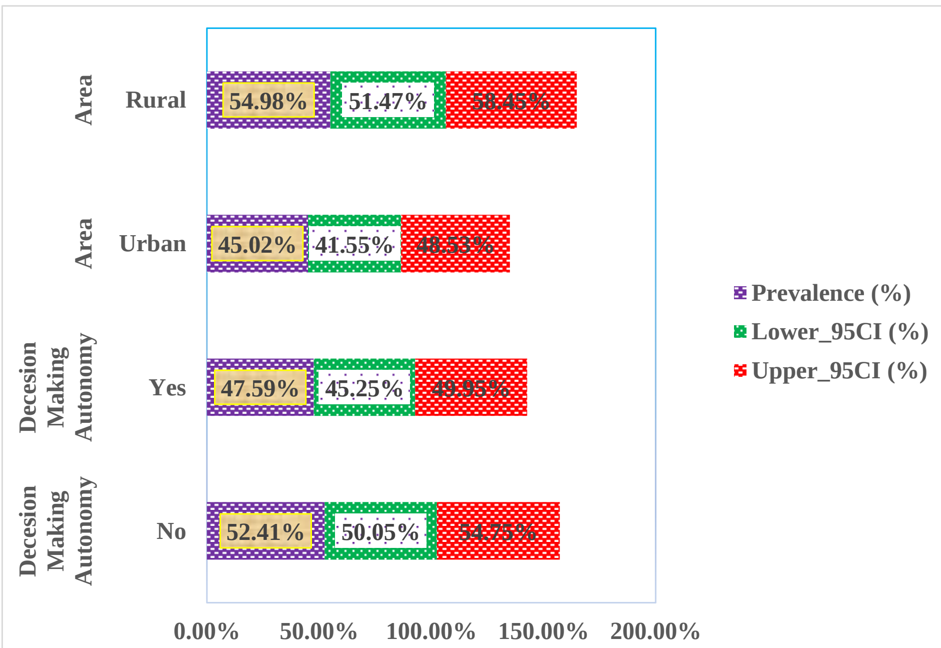

Figure 4: Rural-Urban Inequalities of Decision -Making Autonomy

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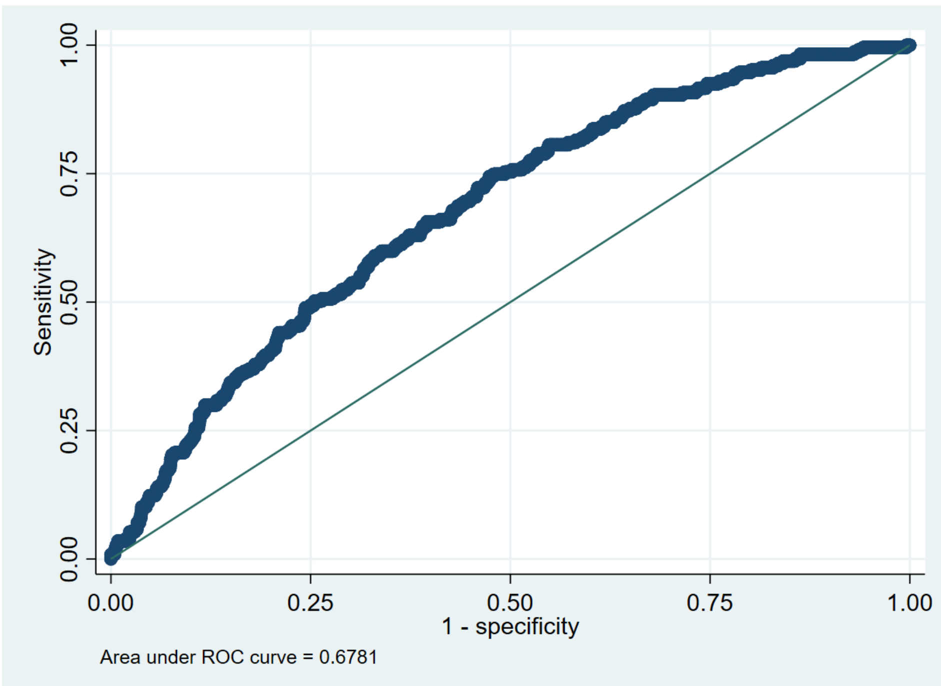

Figure 5:ROC curve showing model performance

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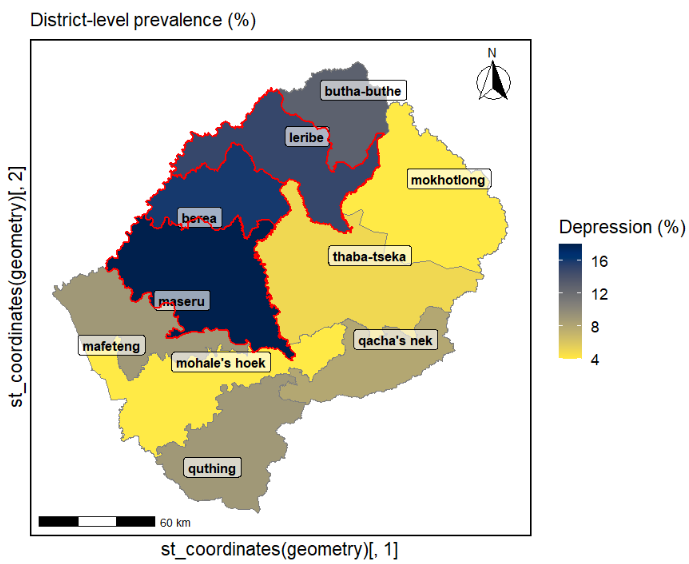

Figure 6: Spatial Inequalities of Depression Prevalences

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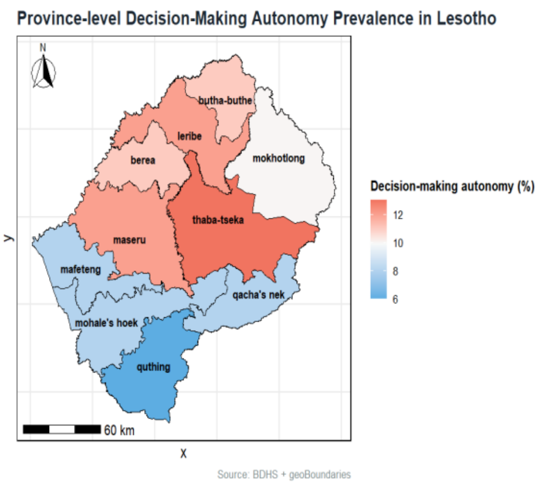

Figure 7: Spatial Inequalities of Decision-making Autonomy

Table 1: Distribution of depressive symptoms (PHQ-9 ≥10) by sociodemographic and reproductive characteristics (N=3,297)

Characteristics	Depressive Symptoms (>=10) %, (95% CI)	
	No	Yes
Overall	92.6 (91.3, 93.7)	7.4 (6.3, 8.7)
Women's age		
15-19	19.9 (18.2, 21.7)	19.6 (13.4, 27.8)
20-24	18.4 (16.7, 20.3)	14.5 (9.3, 21.8)
25-29	13.5 (11.9, 15.3)	16.2 (10.1, 24.9)
30-34	13.6 (12.2, 15.1)	13.2 (8.4, 20.1)
35-39	13.1 (11.6, 14.9)	16.6 (11.1, 24.1)
40-44	11.9 (10.4, 13.6)	13.5(8.9, 20.0)
45-49	9.6 (8.0, 11.5)	6.5 (3.7, 11.1)
Age at first birth		
10-17 (Early)	11.4 (9.6, 13.4)	17.1 (11.6, 24.5)
18-24 (Normal)	44.0 (41.8, 46.3)	45.8 (38.1, 53.7)
≥25 (Late)	44.6 (42.4, 46.8)	37.0 (29.9, 44.8)
Age at first sexual intercourse		
<18	54.4 (51.6, 57.2)	64.0 (56.4, 70.9)
≥18	45.6 (42.8, 48.4)	36.0 (29.1, 43.6)
Age at cohabitation		
<18 (Young)	14.2 (12.4, 16.1)	19.4 (13.9, 26.3)
18–20 (Mild)	20.2 (18.5, 22.0)	17.0 (11.7, 24.0)
≥21 (Older)	65.6 (63.3, 67.9)	63.7 (55.2, 71.3)
Women's education		
Below Primary	24.7 (22.1, 27.4)	28.7 (21.0, 38.0)
Secondary incomplete	59.2 (56.5, 61.9)	54.1 (45.3, 62.6)
Secondary or higher	16.1 (13.3, 19.5)	17.2 (11.4, 25.2)
Maternal Employment		
Not working	61.0 (57.7, 64.2)	61.6 (52.9, 69.6)
Currently working	39.0 (35.8, 42.3)	38.4 (30.4, 47.1)
Adequate ANC Visits		
<4 visits	81.0 (79.1, 82.8)	82.5 (75.5, 87.9)
≥4 visits	19.0 (17.2, 20.9)	17.5 (12.1, 24.5)
Number of children		
None	33.6 (31.5, 35.8)	26.7 (20.4, 34.2)
1-2	48.2 (45.5, 51.0)	48.5 (40.9, 56.1)
3+	18.2 (16.4, 20.0)	24.8 (18.8, 32.1)
Birth Size		
Small	81.5 (79.6, 83.2)	78.7 (71.3, 84.7)
Average or Large	18.5 (16.8, 20.4)	21.3 (15.3, 28.7)
Preterm birth		
Term (>=9 months)	91.8 (90.1, 93.2)	90.9 (84.8, 94.7)

Preterm (<9 months)	8.2 (6.8, 9.9)	9.1 (5.3, 15.2)
Currently Pregnant		
No	97.1 (96.2, 97.7)	98.0 (94.7, 99.2)
Yes	2.9 (2.3, 3.8)	2.0 (0.8, 5.3)
Pregnancy Status		
Single pregnancy	67.7 (65.5, 69.8)	75.3 (68.8, 81.0)
Multiple pregnancy	32.3 (30.2, 34.5)	24.7 (19.2, 31.2)
Pressure to be pregnant		
No	93.3 (91.5, 94.7)	86.3 (76.2, 92.6)
Yes	6.7 (5.3, 8.5)	13.7 (7.4, 23.8)
History of pregnancy loss		
None	89.7 (88.0, 91.1)	84.5 (78.1, 89.3)
One loss	8.2 (6.8, 9.8)	12.6 (8.2, 18.8)
Two or more losses	2.2 (1.6, 3.0)	2.9 (1.3, 6.7)
Residing Status		
Not residing with husband/partner	35.5 (32.0, 39.2)	37.2 (26.6, 49.2)
Residing with husband/partner	64.5 (60.8, 68.0)	62.8 (50.8, 73.4)
Currently Abstaining		
No	93.6 (92.5, 94.5)	94.6 (90.4, 97.1)
Yes	6.4 (5.5, 7.5)	5.4 (2.9, 9.6)
Number of Unions		
Once	60.5 (58.0, 62.9)	57.3 (48.0, 66.2)
More than once	39.5 (37.1, 42.0)	42.7 (33.8, 52.0)
Menstruated last 6 weeks		
No	24.7 (22.8, 26.8)	27.5 (20.6, 35.8)
Yes	75.3 (73.2, 77.2)	72.5 (64.2, 79.4)
Decision Making Autonomy		
No	52.1 (49.7, 54.5)	56.6 (47.0, 65.7)
Yes	47.9 (45.5, 50.3)	43.4 (34.3, 53.0)
IPV Justification		
Does not justify IPV	80.2 (77.8, 82.5)	83.9 (77.7, 88.6)
Justifies IPV	19.8 (17.5, 22.2)	16.1 (11.4, 22.3)
Household Sanitation facility		
Unimproved	35.4 (32.2, 38.8)	30.6 (23.1, 39.3)
Improved	64.6 (61.2, 67.8)	69.4 (60.7, 76.9)
Source of drinking water		
Unimproved source	16.2 (13.7, 19.0)	13.1 (8.8, 19.1)
Improved source	83.8 (81.0, 86.3)	86.9 (80.9, 91.2)
Place of residence		
Urban	44.3 (40.7, 47.9)	54.5 (46.1, 62.7)
Rural	55.7 (52.1, 59.3)	45.5 (37.3, 53.9)
Administrative province		
Butha-Buthe	6.2 (5.3, 7.2)	6.7 (4.5, 9.8)

Leribe	17.8 (15.4, 20.5)	19.5 (14.1, 26.4)
Berea	15.0 (12.0, 18.6)	18.4 (12.3, 26.6)
Maseru	32.8 (28.8, 37.0)	36.8 (28.3, 46.2)
Mafeteng	6.5 (5.5, 7.5)	5.3 (3.5, 8.0)
Mohale’s Hoek	4.7 (4.0, 5.6)	2.2 (1.1, 4.5)
Quthing	3.6 (3.1, 4.3)	3.7 (2.3, 5.8)
Qacha’s Nek	2.9 (2.4, 3.5)	2.9 (1.6, 5.2)
Mokhotlong	4.4 (3.5, 5.4)	1.4 (0.7, 2.7)
Thaba-Tseka	6.2 (5.4, 7.1)	3.1 (1.7, 5.6)
Religion		
Other	65.3 (62.9, 67.6)	63.3 (54.5, 71.4)
Roman Catholic	34.7 (32.4, 37.1)	36.7 (28.6, 45.5)
Currently working		
No	61.0 (57.7, 64.2)	61.6 (52.9, 69.6)
Yes	39.0 (35.8, 42.3)	38.4 (30.4, 47.1)
Wealth Status		
Poor	31.3 (27.6, 35.2)	24.9 (19.0, 31.9)
Middle class	19.2 (16.9, 21.7)	19.1 (13.2, 26.9)
Rich	49.4 (45.2, 53.6)	55.9 (47.1, 64.3)
Household Size		
1-3 members	28.3 (25.6, 31.1)	21.7 (15.9, 28.9)
4 or more members	71.6 (68.8, 74.3)	78.2 (71.0, 84.0)
Household head sex		
Male	57.6 (54.9, 60.2)	55.4 (45.6, 64.8)
Female	42.3 (39.7, 45.0)	44.5 (35.1, 54.3)
Household assets		
No	51.3 (47.3, 55.3)	49.2 (41.6, 56.8)
Yes	48.6 (44.6, 52.6)	50.7 (43.1, 58.3)
Media Access		
No	5.7 (4.77, 6.90)	3.2 (1.79, 5.83)
Yes	94.2 (93.0, 95.2)	96.7 (94.1, 98.2)
Internet use status		
No internet use	19.6 (17.7, 21.6)	22.0 (14.7, 31.4)
Internet use last 12 months	80.3 (78.3, 82.2)	77.9 (68.5, 85.2)
Household materials		
Unimproved	17.0 (14.9, 19.2)	11.5 (7.8, 16.7)
Improved	82.9 (80.7, 85.0)	88.4 (83.2, 92.1)

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Table 2:Associations Between Decision Making Autonomy and Depression Symptoms (N= 3,297, N1 Rural= 2,086, N2 Urban =1,211)

Characteristics	Depression (PHQ-9 ≥10)			
	COR (95% CI), p	AOR (95% CI), p		
Decision Making Autonomy	-	**National Level**	**Rural**	**Urban**
No (Ref)	1	1	1	1
Yes	0.83 0.56-1.23) , 0.368	0.63 (0.39-0.98), 0.048	0.44 (0.27-0.74) , 0.002	0.88 (0.39-1.97) , 0.758

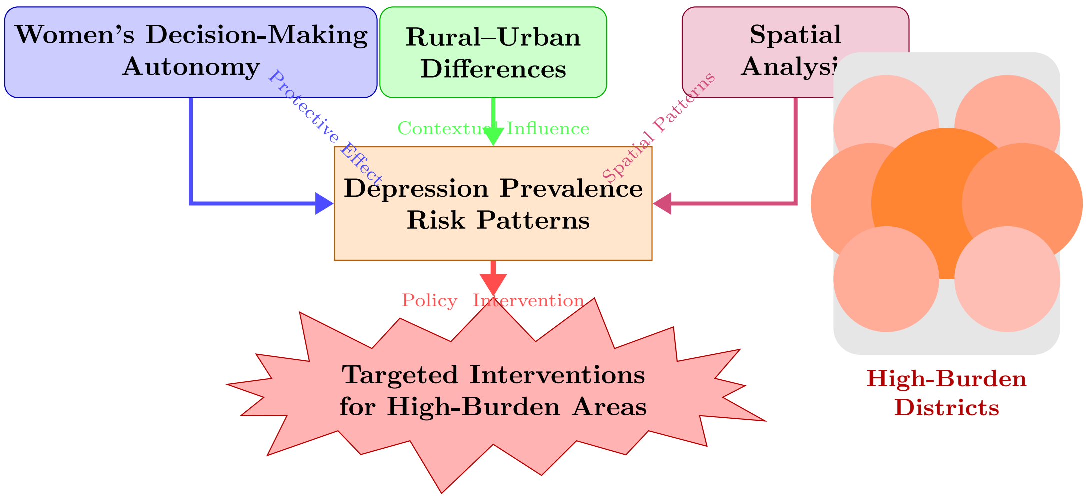
Note: OR = odds ratio; CI = confidence interval; PHQ-9 ≥10 indicates depression; GAD-7 ≥10 indicates anxiety.

Confounders: women age, women education, maternal employment, religion, administrative division, age at first birth, age at cohabitation, number of children, pregnancy status, pregnancy loss, place of residence, toilet facilities, water access, household head sex, household size category, household wealth status, media access, internet use, household assets, and household materials

Graphical abstract

Decision-Making Autonomy and Depression

Rural–Urban and Spatial Disparities in Lesotho



Abstract

Objectives:

To assess the association between depressive symptoms and decision-making autonomy with rural urban inequalities, and mapped province-level spatial disparities in depression prevalence in ever-married women in Lesotho.

Methods:

This study used cross-section data of 3,297 women aged 15–49 years from the 2023–24 Lesotho Demographic and Health Survey (LDHS). Depression was measured using the Patient Health Questionnaire-9 (PHQ-9 ≥10). Women's decision-making autonomy was measured using a composite empowerment index based on participation in five household decision-making domains. Survey-weighted multivariable logistic regression models were fitted at the national level and stratified by place of residence. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC). Spatial analyses were conducted to examine district-level variation in depression prevalence and autonomy.

Results:

Overall, 7.4% (95% CI: 6.3–8.7) of women reported moderate-to-severe depressive symptoms. Women with decision-making autonomy showed lower odds of depression nationally (AOR = 0.63; 95% CI: 0.39–0.98), with a stronger effect in rural areas (AOR = 0.44; 95% CI: 0.27–0.74). Depression prevalence was highest in central–northern districts (Maseru, Berea, Leribe, Butha-Buthe) and lower levels of autonomy highlighted in southern and south-western provinces.

Conclusions:

Women’s decision-making autonomy is significantly associated with lower odds of depressive symptoms in Lesotho, particularly among rural women. The detected rural–urban and spatial inequalities highlight the need for context-specific mental health and reproductive health interventions that incorporate women’s empowerment as a core component.

Keywords: Depression, Decision-making autonomy, Pregnancy loss, Rural–urban disparities, Lesotho, Spatial analysis